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PARALLEL EVAPORATOR COILS FOR SUPERHEAT CONTROL FOR A DEHUMIDIFICATION SYSTEM

Abstract

A dehumidification system includes a primary metering device, a secondary metering device, an intermixed coil unit, a primary evaporator, a secondary condenser, a first modulating valve, and a compressor.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional of U.S. application Ser. No. 18/305,662 filed Apr. 24, 2023 by Scott E. Sloan et al. and entitled “PARALLEL EVAPORATOR COILS FOR SUPERHEAT CONTROL FOR A DEHUMIDIFICATION SYSTEM”, which is a continuation-in-part and claims priority to U.S. Non-provisional application Ser. No. 17/197,781 filed Mar. 10, 2021 by Weizhong Yu et al. and entitled “HEAT MODULATION DEHUMIDIFICATION SYSTEM”, now U.S. Pat. No. 11,668,476 issued Jun. 6, 2023, which claims priority to U.S. Non-provisional application Ser. No. 16/234,052 filed Dec. 27, 2018 by Steven S. Dingle et al. and entitled “SPLIT DEHUMIDIFICATION SYSTEM WITH SECONDARY EVAPORATOR AND CONDENSER COILS”, now U.S. Pat. No. 10,955,148 issued Mar. 23, 2021, which claims priority to U.S. Non-provisional application Ser. No. 15/460,772 filed Mar. 16, 2017 by Dwaine Walter Tucker et al. and entitled “DEHUMIDIFIER WITH SECONDARY EVAPORATOR AND CONDENSER COILS,” now U.S. Pat. No. 10,168,058 issued Jan. 1, 2019, which are all hereby incorporated by reference as if reproduced in their entirety.

TECHNICAL FIELD

[0002] This invention relates generally to dehumidification and more particularly to a dehumidifier with secondary evaporator and condenser coils.

BACKGROUND OF THE INVENTION

[0003] In certain situations, it is desirable to reduce the humidity of air within a structure. For example, in fire and flood restoration applications, it may be desirable to quickly remove water from areas of a damaged structure. To accomplish this, one or more portable dehumidifiers may be placed within the structure to direct dry air toward water-damaged areas. Current dehumidifiers, however, have proven inefficient in various respects.

SUMMARY OF THE INVENTION

[0004] According to embodiments of the present disclosure, disadvantages and problems associated with previous systems may be reduced or eliminated.

[0005] In certain embodiments, a dehumidification system comprises a primary metering device, a secondary metering device, and a superheat control evaporator. The superheat control evaporator is operable to receive a flow of refrigerant from a primary evaporator and to receive an inlet airflow and output a first airflow, the first airflow comprising cooler air than the inlet airflow, the first airflow generated by transferring heat from the inlet airflow to the flow of refrigerant as the inlet airflow passes through the superheat control evaporator. The dehumidification system further comprises a secondary evaporator disposed downstream of and in series with the superheat control evaporator with respect to the airflow. The secondary evaporator is operable to receive the flow of refrigerant from the primary metering device and to receive the first airflow and output a second airflow, the second airflow comprising cooler air than the first airflow, the second airflow generated by transferring heat from the first airflow to the flow of refrigerant as the first airflow passes through the secondary evaporator. The dehumidification system further comprises a primary evaporator operable to receive the flow of refrigerant from a first modulating valve and to receive the second airflow and output a third airflow, the third airflow comprising cooler air than the second airflow, the third airflow generated by transferring heat from the second airflow to the flow of refrigerant as the second airflow passes through the primary evaporator. The dehumidification system further comprises a secondary condenser operable to receive the flow of refrigerant from the first modulating valve and to receive the third airflow and output a fourth airflow.

[0006] The dehumidification system further comprises the first modulating valve operable to receive the flow of refrigerant from the secondary evaporator. During a first mode of operation, the

first modulating valve is operable to direct the flow of refrigerant to the secondary condenser. During a second mode of operation, the first modulating valve is operable direct the flow of refrigerant to the primary evaporator during a second mode of operation, wherein the flow of refrigerant bypasses the secondary condenser. The dehumidification system is further configured to operate in a third mode of operation wherein the first modulating valve is operable to direct a portion of the flow of refrigerant to the secondary condenser and a remaining portion of the flow of refrigerant to the primary evaporator. The dehumidification system further comprises a compressor operable to receive the flow of refrigerant from the superheat control evaporator and discharge the flow of refrigerant at a higher pressure than the flow of refrigerant received at the compressor. The dehumidification system further comprises a primary condenser operable to receive the flow of refrigerant discharged from the compressor. In response to receiving the flow of refrigerant from the compressor, the primary condenser is further operable to output a dischargeable airflow.

[0007] In certain embodiments, a dehumidification system comprises a primary metering device, a secondary metering device, and a superheat control evaporator. The superheat control evaporator is operable to receive a flow of refrigerant from a primary evaporator and to receive a first inlet airflow and output a first airflow, the first airflow comprising cooler air than the first inlet airflow, the first airflow generated by transferring heat from the first inlet airflow to the flow of refrigerant as the first inlet airflow passes through the superheat control evaporator. The dehumidification system further comprises a secondary evaporator disposed in parallel to the superheat control evaporator. The secondary evaporator is operable to receive the flow of refrigerant from the primary metering device and to receive a second inlet airflow and output a second airflow, the second airflow comprising cooler air than the second inlet airflow, the second airflow generated by transferring heat from the second inlet airflow to the flow of refrigerant as the second inlet airflow passes through the secondary evaporator. The dehumidification system further comprises a primary evaporator operable to receive the flow of refrigerant from a first modulating valve and to receive both the first airflow and the second airflow and output a third airflow, the third airflow comprising cooler air than both the first airflow and the second airflow, the third airflow generated by transferring heat from both the first airflow and the second airflow to the flow of refrigerant as both the first airflow and the second airflow pass through the primary evaporator. The dehumidification system further comprises a secondary condenser operable to receive the flow of refrigerant from the first modulating valve and to receive the third airflow and output a fourth airflow.

[0008] The dehumidification system further comprises the first modulating valve operable to receive the flow of refrigerant from the secondary evaporator. During a first mode of operation, the first modulating valve is operable to direct the flow of refrigerant to the secondary condenser. During a second mode of operation, the first modulating valve is operable direct the flow of refrigerant to the primary evaporator during a second mode of operation, wherein the flow of refrigerant bypasses the secondary condenser. The dehumidification system is further configured to operate in a third mode of operation wherein the first modulating valve is operable to direct a portion of the flow of refrigerant to the secondary condenser and a remaining portion of the flow of refrigerant to the primary evaporator. The dehumidification system further comprises a compressor operable to receive the flow of refrigerant from the superheat control evaporator and discharge the flow of refrigerant at a higher pressure than the flow of refrigerant received at the compressor. The dehumidification system further comprises a primary condenser operable to receive the flow of refrigerant discharged from the compressor. In response to receiving the flow of refrigerant from the compressor, the primary condenser is further operable to output a dischargeable airflow.

[0009] In certain embodiments, a dehumidification system comprises a primary metering device, a secondary metering device, and an intermixed coil unit. The intermixed coil unit is operable to receive an inlet airflow and output a first airflow, the first airflow comprising cooler air than the first inlet airflow, the first airflow generated by transferring heat from the inlet airflow to a flow of refrigerant within a superheat control evaporator and a secondary evaporator as the inlet airflow

passes through both the superheat control evaporator and the secondary evaporator. The intermixed coil unit comprises the superheat control evaporator and the secondary evaporator. The superheat control evaporator is operable to receive a flow of refrigerant from a primary evaporator, and the secondary evaporator is operable to receive the flow of refrigerant from the primary metering device. The dehumidification system further comprises a primary evaporator operable to receive the flow of refrigerant from a first modulating valve and to receive the first airflow and output a second airflow, the second airflow comprising cooler air than the first airflow, the second airflow generated by transferring heat from the first airflow to the flow of refrigerant as the first airflow pass through the primary evaporator. The dehumidification system further comprises a secondary condenser operable to receive the flow of refrigerant from the first modulating valve and to receive the second airflow and output a third airflow.

[0010] The dehumidification system further comprises the first modulating valve operable to receive the flow of refrigerant from the secondary evaporator. During a first mode of operation, the first modulating valve is operable to direct the flow of refrigerant to the secondary condenser. During a second mode of operation, the first modulating valve is operable direct the flow of refrigerant to the primary evaporator during a second mode of operation, wherein the flow of refrigerant bypasses the secondary condenser. The dehumidification system is further configured to operate in a third mode of operation wherein the first modulating valve is operable to direct a portion of the flow of refrigerant to the secondary condenser and a remaining portion of the flow of refrigerant to the primary evaporator. The dehumidification system further comprises a compressor operable to receive the flow of refrigerant from the superheat control evaporator and discharge the flow of refrigerant at a higher pressure than the flow of refrigerant received at the compressor. The dehumidification system further comprises a primary condenser operable to receive the flow of refrigerant discharged from the compressor. In response to receiving the flow of refrigerant from the compressor, the primary condenser is further operable to output a dischargeable airflow.

[0011] Certain embodiments of the present disclosure may provide one or more technical advantages. The advantages with these embodiments include modulating the amount of sensible heat to latent heat. For example, adjusting the ratio of sensible to latent heat can further decrease the temperature of the surrounding airflow or increase the amount of water removed from the airflow. The secondary condenser may be isolated, wherein energy is not recovered by the airflow via refrigerant flowing through the secondary condenser, thereby providing an air cooling operation rather than dehumidification. Further, modulation is based off controlling the superheat from one or more evaporator coils of the dehumidification system. The addition of another evaporator coil upstream of the primary evaporator provides faster and more efficient control of operations of the dehumidification system. For example, providing increased superheat control can prevent the cooling of an airflow already determined to be at a control or set temperature.

[0012] In further examples, certain embodiments include two evaporators, two condensers, and two metering devices that utilize a closed refrigeration loop. This configuration causes part of the refrigerant within the system to evaporate and condense twice in one refrigeration cycle, thereby increasing the compressor capacity over typical systems without adding any additional power to the compressor. This, in turn, increases the overall efficiency of the system by providing more dehumidification per kilowatt of power used. The lower humidity of the output airflow may allow for increased drying potential, which may be beneficial in certain applications (e.g., fire and flood restoration).

[0013] Certain embodiments of the present disclosure may include some, all, or none of the above advantages. One or more other technical advantages may be readily apparent to those skilled in the art from the figures, descriptions, and claims included herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] To provide a more complete understanding of the present invention and the features and advantages thereof, reference is made to the following description taken in conjunction with the accompanying drawings, in which:

[0015] FIG. 1 illustrates an example split system for reducing the humidity of air within a structure, according to certain embodiments;

[0016] FIG. 2 illustrates an example portable system for reducing the humidity of air within a structure, according to certain embodiments;

[0017] FIGS. 3 and 4 illustrate an example dehumidification system that may be used by the systems of FIGS. 1 and 2 to reduce the humidity of air within a structure, according to certain embodiments;

[0018] FIG. 5 illustrates an example dehumidification method that may be used by the systems of FIGS. 1 and 2 to reduce the humidity of air within a structure, according to certain embodiments;

[0019] FIGS. 6A and 6B illustrate an example air conditioning and dehumidification system, according to certain embodiments;

[0020] FIG. 7 illustrates an example condenser system for use in the system described herein, according to certain embodiments;

[0021] FIGS. 8A, 8B, and 8C illustrate an example air conditioning and dehumidification system, according to certain embodiments;

[0022] FIGS. 9 and 10 illustrate examples of single coil packs for use in the system described herein, according to certain embodiments;

[0023] FIGS. 11, 12, 13, and 14 illustrate an example of a primary evaporator comprising three circuits for use in the system described herein, according to certain embodiments;

[0024] FIGS. 15A and 15B illustrate an example dehumidification system with a liquid cooled condenser, according to certain embodiments; and

[0025] FIGS. 16A, 16B, 16C, and 16D illustrate an example dehumidification system with a modulating valve, according to certain embodiments.

[0026] FIGS. 17A, 17B, and 17C illustrate an example dehumidification system with a superheat control evaporator, according to certain embodiments.

DETAILED DESCRIPTION OF THE DRAWINGS

[0027] In certain situations, it is desirable to reduce the humidity of air within a structure. For example, in fire and flood restoration applications, it may be desirable to remove water from a damaged structure by placing one or more portable dehumidifiers unit within the structure. As another example, in areas that experience weather with high humidity levels, or in buildings where low humidity levels are required (e.g., libraries), it may be desirable to install a dehumidification unit within a central air conditioning system. Furthermore, it may be necessary to hold a desired humidity level in some commercial applications. Current dehumidifiers, however, have proven inadequate or inefficient in various respects.

[0028] To address the inefficiencies and other issues with current dehumidification systems, the disclosed embodiments provide a dehumidification system that includes a secondary evaporator and a secondary condenser, which causes part of the refrigerant within the multi-stage system to evaporate and condense twice in one refrigeration cycle. This increases the compressor capacity over typical systems without adding any additional power to the compressor. This, in turn, increases the overall efficiency of the system by providing more dehumidification per kilowatt of power used.

[0029] FIG. 1 illustrates an example dehumidification system **100** for supplying dehumidified air **106** to a structure **102**, according to certain embodiments. Dehumidification system **100** includes an evaporator system **104** located within structure **102**. Structure **102** may include all or a portion of a building or other suitable enclosed space, such as an apartment building, a hotel, an office space, a

commercial building, or a private dwelling (e.g., a house). Evaporator system **104** receives inlet air **101** from within structure **102**, reduces the moisture in received inlet air **101**, and supplies dehumidified air **106** back to structure **102**. Evaporator system **104** may distribute dehumidified air **106** throughout structure **102** via air ducts, as illustrated.

[0030] In general, dehumidification system **100** is a split system wherein evaporator system **104** is coupled to a remote condenser system **108** that is located external to structure **102**. Remote condenser system **108** may include a condenser unit **112** and a compressor unit **114** that facilitate the functions of evaporator system **104** by processing a flow of refrigerant as part of a refrigeration cycle. The flow of refrigerant may include any suitable cooling material, such as R410a refrigerant. In certain embodiments, compressor unit **114** may receive the flow of refrigerant vapor from evaporator system **104** via a refrigerant line **116**. Compressor unit **114** may pressurize the flow of refrigerant, thereby increasing the temperature of the refrigerant. The speed of the compressor may be modulated to effectuate desired operating characteristics. Condenser unit **112** may receive the pressurized flow of refrigerant vapor from compressor unit **114** and cool the pressurized refrigerant by facilitating heat transfer from the flow of refrigerant to the ambient air exterior to structure **102**. In certain embodiments, remote condenser system **108** may utilize a heat exchanger, such as a microchannel heat exchanger to remove heat from the flow of refrigerant. Remote condenser system **108** may include a fan that draws ambient air from outside structure **102** for use in cooling the flow of refrigerant. In certain embodiments, the speed of this fan is modulated to effectuate desired operating characteristics. An illustrative embodiment of an example condenser system is shown, for example, in FIG. 7 (described in further detail below).

[0031] After being cooled and condensed to liquid by condenser unit **112**, the flow of refrigerant may travel by a refrigerant line **118** to evaporator system **104**. In certain embodiments, the flow of refrigerant may be received by an expansion device (described in further detail below) that reduces the pressure of the flow of refrigerant, thereby reducing the temperature of the flow of refrigerant. An evaporator unit (described in further detail below) of evaporator system **104** may receive the flow of refrigerant from the expansion device and use the flow of refrigerant to dehumidify and cool an incoming airflow. The flow of refrigerant may then flow back to remote condenser system **108** and repeat this cycle.

[0032] In certain embodiments, evaporator system **104** may be installed in series with an air mover. An air mover may include a fan that blows air from one location to another. An air mover may facilitate distribution of outgoing air from evaporator system **104** to various parts of structure **102**. An air mover and evaporator system **104** may have separate return inlets from which air is drawn. In certain embodiments, outgoing air from evaporator system **104** may be mixed with air produced by another component (e.g., an air conditioner) and blown through air ducts by the air mover. In other embodiments, evaporator system **104** may perform both cooling and dehumidifying and thus may be used without a conventional air conditioner.

[0033] Although a particular implementation of dehumidification system **100** is illustrated and primarily described, the present disclosure contemplates any suitable implementation of dehumidification system **100**, according to particular needs. Moreover, although various components of dehumidification system **100** have been depicted as being located at particular positions, the present disclosure contemplates those components being positioned at any suitable location, according to particular needs.

[0034] FIG. 2 illustrates an example portable dehumidification system **200** for reducing the humidity of air within structure **102**, according to certain embodiments of the present disclosure. Dehumidification system **200** may be positioned anywhere within structure **102** in order to direct dehumidified air **106** towards areas that require dehumidification (e.g., water-damaged areas). In general, dehumidification system **200** receives inlet airflow **101**, removes water from the inlet airflow **101**, and discharges dehumidified air **106** air back into structure **102**. In certain embodiments, structure **102** includes a space that has suffered water damage (e.g., as a result of a

flood or fire). In order to restore the water-damaged structure **102**, one or more dehumidification systems **200** may be strategically positioned within structure **102** in order to quickly reduce the humidity of the air within the structure **102** and thereby dry the portions of structure **102** that suffered water damage.

[0035] Although a particular implementation of portable dehumidification system **200** is illustrated and primarily described, the present disclosure contemplates any suitable implementation of portable dehumidification system **200**, according to particular needs. Moreover, although various components of portable dehumidification system **200** have been depicted as being located at particular positions within structure **102**, the present disclosure contemplates those components being positioned at any suitable location, according to particular needs.

[0036] FIGS. **3** and **4** illustrate an example dehumidification system **300** that may be used by dehumidification system **100** and portable dehumidification system **200** of FIGS. **1** and **2** to reduce the humidity of air within structure **102**. Dehumidification system **300** includes a primary evaporator **310**, a primary condenser **330**, a secondary evaporator **340**, a secondary condenser **320**, a compressor **360**, a primary metering device **380**, a secondary metering device **390**, and a fan **370**. In some embodiments, dehumidification system **300** may additionally include a sub-cooling coil **350**. In certain embodiments, sub-cooling coil **350** and primary condenser **330** are combined into a single coil. A flow of refrigerant **305** is circulated through dehumidification system **300** as illustrated. In general, dehumidification system **300** receives inlet airflow **101**, removes water from inlet airflow **101**, and discharges dehumidified air **106**. Water is removed from inlet air **101** using a refrigeration cycle of flow of refrigerant **305**. By including secondary evaporator **340** and secondary condenser **320**, however, dehumidification system **300** causes at least part of the flow of refrigerant **305** to evaporate and condense twice in a single refrigeration cycle. This increases the refrigeration capacity over typical systems without adding any additional power to the compressor, thereby increasing the overall dehumidification efficiency of the system.

[0037] In general, dehumidification system **300** attempts to match the saturating temperature of secondary evaporator **340** to the saturating temperature of secondary condenser **320**. The saturating temperature of secondary evaporator **340** and secondary condenser **320** generally is controlled according to the equation: (temperature of inlet air **101**+temperature of second airflow **315**)/2. As the saturating temperature of secondary evaporator **340** is lower than inlet air **101**, evaporation happens in secondary evaporator **340**. As the saturating temperature of secondary condenser **320** is higher than second airflow **315**, condensation happens in the secondary condenser **320**. The amount of refrigerant **305** evaporating in secondary evaporator **340** is substantially equal to that condensing in secondary condenser **320**.

[0038] Primary evaporator **310** receives flow of refrigerant **305** from secondary metering device **390** and outputs flow of refrigerant **305** to compressor **360**. Primary evaporator **310** may be any type of coil (e.g., fin tube, micro channel, etc.). Primary evaporator **310** receives first airflow **345** from secondary evaporator **340** and outputs second airflow **315** to secondary condenser **320**. Second airflow **315**, in general, is at a cooler temperature than first airflow **345**. To cool incoming first airflow **345**, primary evaporator **310** transfers heat from first airflow **345** to flow of refrigerant **305**, thereby causing flow of refrigerant **305** to evaporate at least partially from liquid to gas. This transfer of heat from first airflow **345** to flow of refrigerant **305** also removes water from first airflow **345**.

[0039] Secondary condenser **320** receives flow of refrigerant **305** from secondary evaporator **340** and outputs flow of refrigerant **305** to secondary metering device **390**. Secondary condenser **320** may be any type of coil (e.g., fin tube, micro channel, etc.). Secondary condenser **320** receives second airflow **315** from primary evaporator **310** and outputs third airflow **325**. Third airflow **325** is, in general, warmer and drier (i.e., the dew point will be the same but relative humidity will be lower) than second airflow **315**. Secondary condenser **320** generates third airflow **325** by transferring heat from flow of refrigerant **305** to second airflow **315**, thereby causing flow of

refrigerant **305** to condense at least partially from gas to liquid.

[0040] Primary condenser **330** receives flow of refrigerant **305** from compressor **360** and outputs flow of refrigerant **305** to either primary metering device **380** or sub-cooling coil **350**. Primary condenser **330** may be any type of coil (e.g., fin tube, micro channel, etc.). Primary condenser **330** receives either third airflow **325** or fourth airflow **355** and outputs dehumidified air **106**.

Dehumidified air **106** is, in general, warmer and drier (i.e., have a lower relative humidity) than third airflow **325** and fourth airflow **355**. Primary condenser **330** generates dehumidified air **106** by transferring heat from flow of refrigerant **305**, thereby causing flow of refrigerant **305** to condense at least partially from gas to liquid. In some embodiments, primary condenser **330** completely condenses flow of refrigerant **305** to a liquid (i.e., 100% liquid). In other embodiments, primary condenser **330** partially condenses flow of refrigerant **305** to a liquid (i.e., less than 100% liquid). In certain embodiments, as shown in FIG. 4, a portion of primary condenser **330** receives a separate airflow in addition to airflow **101**. For example, the right-most edge of primary condenser **330** of FIG. 4 extends beyond, or overhangs, the right-most edges of secondary evaporator **340**, primary evaporator **310**, secondary condenser **320**, and sub-cooling coil **350**. This overhanging portion of primary condenser **330** may receive an additional separate airflow.

[0041] Secondary evaporator **340** receives flow of refrigerant **305** from primary metering device **380** and outputs flow of refrigerant **305** to secondary condenser **320**. Secondary evaporator **340** may be any type of coil (e.g., fin tube, micro channel, etc.). Secondary evaporator **340** receives inlet air **101** and outputs first airflow **345** to primary evaporator **310**. First airflow **345**, in general, is at a cooler temperature than inlet air **101**. To cool incoming inlet air **101**, secondary evaporator **340** transfers heat from inlet air **101** to flow of refrigerant **305**, thereby causing flow of refrigerant **305** to evaporate at least partially from liquid to gas.

[0042] Sub-cooling coil **350**, which is an optional component of dehumidification system **300**, sub-cools the liquid refrigerant **305** as it leaves primary condenser **330**. This, in turn, supplies primary metering device **380** with a liquid refrigerant that is up to 30 degrees (or more) cooler than before it enters sub-cooling coil **350**. For example, if flow of refrigerant **305** entering sub-cooling coil **350** is 340 psig/105° F./60% vapor, flow of refrigerant **305** may be 340 psig/80° F./0% vapor as it leaves sub-cooling coil **350**. The sub-cooled refrigerant **305** has a greater heat enthalpy factor as well as a greater density, which results in reduced cycle times and frequency of the evaporation cycle of flow of refrigerant **305**. This results in greater efficiency and less energy use of dehumidification system **300**. Embodiments of dehumidification system **300** may or may not include a sub-cooling coil **350**. For example, embodiments of dehumidification system **300** utilized within portable dehumidification system **200** that have a micro-channel condenser **330** or **320** may include a sub-cooling coil **350**, while embodiments of dehumidification system **300** that utilize another type of condenser **330** or **320** may not include a sub-cooling coil **350**. As another example, dehumidification system **300** utilized within a split system such as dehumidification system **100** may not include a sub-cooling coil **350**.

[0043] Compressor **360** pressurizes flow of refrigerant **305**, thereby increasing the temperature of refrigerant **305**. For example, if flow of refrigerant **305** entering compressor **360** is 128 psig/52° F./100% vapor, flow of refrigerant **305** may be 340 psig/150° F./100% vapor as it leaves compressor **360**. Compressor **360** receives flow of refrigerant **305** from primary evaporator **310** and supplies the pressurized flow of refrigerant **305** to primary condenser **330**.

[0044] Fan **370** may include any suitable components operable to draw inlet air **101** into dehumidification system **300** and through secondary evaporator **340**, primary evaporator **310**, secondary condenser **320**, sub-cooling coil **350**, and primary condenser **330**. Fan **370** may be any type of air mover (e.g., axial fan, forward inclined impeller, and backward inclined impeller, etc.). For example, fan **370** may be a backward inclined impeller positioned adjacent to primary condenser **330** as illustrated in FIG. 3. While fan **370** is depicted in FIG. 3 as being located adjacent to primary condenser **330**, it should be understood that fan **370** may be located anywhere

along the airflow path of dehumidification system **300**. For example, fan **370** may be positioned in the airflow path of any one of airflows **101**, **345**, **315**, **325**, **355**, or **106**. Moreover, dehumidification system **300** may include one or more additional fans positioned within any one or more of these airflow paths.

[0045] Primary metering device **380** and secondary metering device **390** are any appropriate type of metering/expansion device. In some embodiments, primary metering device **380** is a thermostatic expansion valve (TXV) and secondary metering device **390** is a fixed orifice device (or vice versa). In certain embodiments, metering devices **380** and **390** remove pressure from flow of refrigerant **305** to allow expansion or change of state from a liquid to a vapor in evaporators **310** and **340**. The high-pressure liquid (or mostly liquid) refrigerant entering metering devices **380** and **390** is at a higher temperature than the liquid refrigerant **305** leaving metering devices **380** and **390**. For example, if flow of refrigerant **305** entering primary metering device **380** is 340 psig/80° F./0% vapor, flow of refrigerant **305** may be 196 psig/68° F./5% vapor as it leaves primary metering device **380**. As another example, if flow of refrigerant **305** entering secondary metering device **390** is 196 psig/68° F./4% vapor, flow of refrigerant **305** may be 128 psig/44° F./14% vapor as it leaves secondary metering device **390**.

[0046] Refrigerant **305** may be any suitable refrigerant such as R410a. In general, dehumidification system **300** utilizes a closed refrigeration loop of refrigerant **305** that passes from compressor **360** through primary condenser **330**, (optionally) sub-cooling coil **350**, primary metering device **380**, secondary evaporator **340**, secondary condenser **320**, secondary metering device **390**, and primary evaporator **310**. Compressor **360** pressurizes flow of refrigerant **305**, thereby increasing the temperature of refrigerant **305**. Primary and secondary condensers **330** and **320**, which may include any suitable heat exchangers, cool the pressurized flow of refrigerant **305** by facilitating heat transfer from the flow of refrigerant **305** to the respective airflows passing through them (i.e., fourth airflow **355** and second airflow **315**). The cooled flow of refrigerant **305** leaving primary and secondary condensers **330** and **320** may enter a respective expansion device (i.e., primary metering device **380** and secondary metering device **390**) that is operable to reduce the pressure of flow of refrigerant **305**, thereby reducing the temperature of flow of refrigerant **305**. Primary and secondary evaporators **310** and **340**, which may include any suitable heat exchanger, receive flow of refrigerant **305** from secondary metering device **390** and primary metering device **380**, respectively. Primary and secondary evaporators **310** and **340** facilitate the transfer of heat from the respective airflows passing through them (i.e., inlet air **101** and first airflow **345**) to flow of refrigerant **305**. Flow of refrigerant **305**, after leaving primary evaporator **310**, passes back to compressor **360**, and the cycle is repeated.

[0047] In certain embodiments, the above-described refrigeration loop may be configured such that evaporators **310** and **340** operate in a flooded state. In other words, flow of refrigerant **305** may enter evaporators **310** and **340** in a liquid state, and a portion of flow of refrigerant **305** may still be in a liquid state as it exits evaporators **310** and **340**. Accordingly, the phase change of flow of refrigerant **305** (liquid to vapor as heat is transferred to flow of refrigerant **305**) occurs across evaporators **310** and **340**, resulting in nearly constant pressure and temperature across the entire evaporators **310** and **340** (and, as a result, increased cooling capacity).

[0048] In operation of example embodiments of dehumidification system **300**, inlet air **101** may be drawn into dehumidification system **300** by fan **370**. Inlet air **101** passes through secondary evaporator **340** in which heat is transferred from inlet air **101** to the cool flow of refrigerant **305** passing through secondary evaporator **340**. As a result, inlet air **101** may be cooled. As an example, if inlet air **101** is 80° F./60% humidity, secondary evaporator **340** may output first airflow **345** at 70° F./84% humidity. This may cause flow of refrigerant **305** to partially vaporize within secondary evaporator **340**. For example, if flow of refrigerant **305** entering secondary evaporator **340** is 196 psig/68° F./5% vapor, flow of refrigerant **305** may be 196 psig/68° F./38% vapor as it leaves secondary evaporator **340**.

[0049] The cooled inlet air **101** leaves secondary evaporator **340** as first airflow **345** and enters primary evaporator **310**. Like secondary evaporator **340**, primary evaporator **310** transfers heat from first airflow **345** to the cool flow of refrigerant **305** passing through primary evaporator **310**. As a result, first airflow **345** may be cooled to or below its dew point temperature, causing moisture in first airflow **345** to condense (thereby reducing the absolute humidity of first airflow **345**). As an example, if first airflow **345** is 70° F./84% humidity, primary evaporator **310** may output second airflow **315** at 54° F./98% humidity. This may cause flow of refrigerant **305** to partially or completely vaporize within primary evaporator **310**. For example, if flow of refrigerant **305** entering primary evaporator **310** is 128 psig/44° F./14% vapor, flow of refrigerant **305** may be 128 psig/52° F./100% vapor as it leaves primary evaporator **310**. In certain embodiments, the liquid condensate from first airflow **345** may be collected in a drain pan connected to a condensate reservoir, as illustrated in FIG. 4. Additionally, the condensate reservoir may include a condensate pump that moves collected condensate, either continually or at periodic intervals, out of dehumidification system **300** (e.g., via a drain hose) to a suitable drainage or storage location.

[0050] The cooled first airflow **345** leaves primary evaporator **310** as second airflow **315** and enters secondary condenser **320**. Secondary condenser **320** facilitates heat transfer from the hot flow of refrigerant **305** passing through the secondary condenser **320** to second airflow **315**. This reheats second airflow **315**, thereby decreasing the relative humidity of second airflow **315**. As an example, if second airflow **315** is 54° F./98% humidity, secondary condenser **320** may output third airflow **325** at 65° F./68% humidity. This may cause flow of refrigerant **305** to partially or completely condense within secondary condenser **320**. For example, if flow of refrigerant **305** entering secondary condenser **320** is 196 psig/68° F./38% vapor, flow of refrigerant **305** may be 196 psig/68° F./4% vapor as it leaves secondary condenser **320**.

[0051] In some embodiments, the dehumidified second airflow **315** leaves secondary condenser **320** as third airflow **325** and enters primary condenser **330**. Primary condenser **330** facilitates heat transfer from the hot flow of refrigerant **305** passing through the primary condenser **330** to third airflow **325**. This further heats third airflow **325**, thereby further decreasing the relative humidity of third airflow **325**. As an example, if third airflow **325** is 65° F./68% humidity, secondary condenser **320** may output dehumidified air **106** at 102° F./19% humidity. This may cause flow of refrigerant **305** to partially or completely condense within primary condenser **330**. For example, if flow of refrigerant **305** entering primary condenser **330** is 340 psig/150° F./100% vapor, flow of refrigerant **305** may be 340 psig/105° F./60% vapor as it leaves primary condenser **330**.

[0052] As described above, some embodiments of dehumidification system **300** may include a sub-cooling coil **350** in the airflow between secondary condenser **320** and primary condenser **330**. Sub-cooling coil **350** facilitates heat transfer from the hot flow of refrigerant **305** passing through sub-cooling coil **350** to third airflow **325**. This further heats third airflow **325**, thereby further decreasing the relative humidity of third airflow **325**. As an example, if third airflow **325** is 65° F./68% humidity, sub-cooling coil **350** may output fourth airflow **355** at 81° F./37% humidity. This may cause flow of refrigerant **305** to partially or completely condense within sub-cooling coil **350**. For example, if flow of refrigerant **305** entering sub-cooling coil **350** is 340 psig/150° F./60% vapor, flow of refrigerant **305** may be 340 psig/80° F./0% vapor as it leaves sub-cooling coil **350**.

[0053] Some embodiments of dehumidification system **300** may include a controller that may include one or more computer systems at one or more locations. Each computer system may include any appropriate input devices (such as a keypad, touch screen, mouse, or other device that can accept information), output devices, mass storage media, or other suitable components for receiving, processing, storing, and communicating data. Both the input devices and output devices may include fixed or removable storage media such as a magnetic computer disk, CD-ROM, or other suitable media to both receive input from and provide output to a user. Each computer system may include a personal computer, workstation, network computer, kiosk, wireless data port, personal data assistant (PDA), one or more processors within these or other devices, or any other

suitable processing device. In short, the controller may include any suitable combination of software, firmware, and hardware.

[0054] The controller may additionally include one or more processing modules. Each processing module may each include one or more microprocessors, controllers, or any other suitable computing devices or resources and may work, either alone or with other components of dehumidification system **300**, to provide a portion or all of the functionality described herein. The controller may additionally include (or be communicatively coupled to via wireless or wireline communication) computer memory. The memory may include any memory or database module and may take the form of volatile or non-volatile memory, including, without limitation, magnetic media, optical media, random access memory (RAM), read-only memory (ROM), removable media, or any other suitable local or remote memory component.

[0055] Although particular implementations of dehumidification system **300** are illustrated and primarily described, the present disclosure contemplates any suitable implementation of dehumidification system **300**, according to particular needs. Moreover, although various components of dehumidification system **300** have been depicted as being located at particular positions and relative to one another, the present disclosure contemplates those components being positioned at any suitable location, according to particular needs.

[0056] FIG. **5** illustrates an example dehumidification method **500** that may be used by dehumidification system **100** and portable dehumidification system **200** of FIGS. **1** and **2** to reduce the humidity of air within structure **102**. Method **500** may begin in step **510** where a secondary evaporator receives an inlet airflow and outputs a first airflow. In some embodiments, the secondary evaporator is secondary evaporator **340**. In some embodiments, the inlet airflow is inlet air **101** and the first airflow is first airflow **345**. In some embodiments, the secondary evaporator of step **510** receives a flow of refrigerant from a primary metering device such as primary metering device **380** and supplies the flow of refrigerant (in a changed state) to a secondary condenser such as secondary condenser **320**. In some embodiments, the flow of refrigerant of method **500** is flow of refrigerant **305** described above.

[0057] At step **520**, a primary evaporator receives the first airflow of step **510** and outputs a second airflow. In some embodiments, the primary evaporator is primary evaporator **310** and the second airflow is second airflow **315**. In some embodiments, the primary evaporator of step **520** receives the flow of refrigerant from a secondary metering device such as secondary metering device **390** and supplies the flow of refrigerant (in a changed state) to a compressor such as compressor **360**.

[0058] At step **530**, a secondary condenser receives the second airflow of step **520** and outputs a third airflow. In some embodiments, the secondary condenser is secondary condenser **320** and the third airflow is third airflow **325**. In some embodiments, the secondary condenser of step **530** receives a flow of refrigerant from the secondary evaporator of step **510** and supplies the flow of refrigerant (in a changed state) to a secondary metering device such as secondary metering device **390**.

[0059] At step **540**, a primary condenser receives the third airflow of step **530** and outputs a dehumidified airflow. In some embodiments, the primary condenser is primary condenser **330** and the dehumidified airflow is dehumidified air **106**. In some embodiments, the primary condenser of step **540** receives a flow of refrigerant from the compressor of step **520** and supplies the flow of refrigerant (in a changed state) to the primary metering device of step **510**. In alternate embodiments, the primary condenser of step **540** supplies the flow of refrigerant (in a changed state) to a sub-cooling coil such as sub-cooling coil **350** which in turn supplies the flow of refrigerant (in a changed state) to the primary metering device of step **510**.

[0060] At step **550**, a compressor receives the flow of refrigerant from the primary evaporator of step **520** and provides the flow of refrigerant (in a changed state) to the primary condenser of step **540**. After step **550**, method **500** may end.

[0061] Particular embodiments may repeat one or more steps of method **500** of FIG. **5**, where

appropriate. Although this disclosure describes and illustrates particular steps of the method of FIG. 5 as occurring in a particular order, this disclosure contemplates any suitable steps of the method of FIG. 5 occurring in any suitable order. Moreover, although this disclosure describes and illustrates an example dehumidification method for reducing the humidity of air within a structure including the particular steps of the method of FIG. 5, this disclosure contemplates any suitable method for reducing the humidity of air within a structure including any suitable steps, which may include all, some, or none of the steps of the method of FIG. 5, where appropriate. Furthermore, although this disclosure describes and illustrates particular components, devices, or systems carrying out particular steps of the method of FIG. 5, this disclosure contemplates any suitable combination of any suitable components, devices, or systems carrying out any suitable steps of the method of FIG. 5.

[0062] While the example method of FIG. 5 is described at times above with respect to dehumidification system **300** of FIG. 3, it should be understood that the same or similar methods can be carried out using any of the dehumidification systems described herein, including dehumidification systems **600** and **800** of FIGS. 6A-6B and 8 (described below). Moreover, it should be understood that, with respect to the example method of FIG. 5, reference to an evaporator or condenser can refer to an evaporator portion or condenser portion of a single coil pack operable to perform the functions of these components, for example, as described above with respect to examples of FIGS. 9 and 10.

[0063] FIGS. 6A and 6B illustrate an example air conditioning and dehumidification system **600** that may be used in accordance with split dehumidification system **100** of FIG. 1 to reduce the humidity of air within structure **102**. Dehumidification system **600** includes a dehumidification unit **602**, which is generally indoors, and a condenser system **604** (e.g., condenser system **108** of FIG. 1). As illustrated in FIG. 6A, dehumidification unit **602** includes a primary evaporator **610**, a secondary evaporator **640**, a secondary condenser **620**, a primary metering device **680**, a secondary metering device **690**, and a first fan **670**, while condenser system **604** includes a primary condenser **630**, a compressor **660**, an optional sub-cooling coil **650** and a second fan **695**. In the embodiment illustrated in FIG. 6B, the compressor **660** may be disposed within the dehumidification unit **602** rather than disposed within the condenser system **604**.

[0064] With reference to both FIGS. 6A and 6B, a flow of refrigerant **605** is circulated through dehumidification system **600** as illustrated. In general, dehumidification unit **602** receives inlet airflow **601**, removes water from inlet airflow **601**, and discharges dehumidified air **625** into a conditioned space. Water is removed from inlet air **601** using a refrigeration cycle of flow of refrigerant **605**. The flow of refrigerant **605** through system **600** of FIGS. 6A AND 6B proceeds in a similar manner to that of the flow of refrigerant **305** through dehumidification system **300** of FIG. 3. However, the path of airflow through system **600** is different than that through system **300**, as described herein. By including secondary evaporator **640** and secondary condenser **620**, however, dehumidification system **600** causes at least part of the flow of refrigerant **605** to evaporate and condense twice in a single refrigeration cycle. This increases refrigerating capacity over typical systems without requiring any additional power to the compressor, thereby increasing the overall efficiency of the system.

[0065] The split configuration of system **600**, which includes dehumidification unit **602** and condenser system **604**, allows heat from the cooling and dehumidification process to be rejected outdoors or to an unconditioned space (e.g., external to a space being dehumidified). This allows dehumidification system **600** to have a similar footprint to that of typical central air conditioning systems or heat pumps. In general, the temperature of third airflow **625** output to the conditioned space from system **600** is significantly decreased compared to that of airflow **106** output from system **300** of FIG. 3. Thus, the configuration of system **600** allows dehumidified air to be provided to the conditioned space at a decreased temperature. Accordingly, system **600** may perform functions of both a dehumidifier (dehumidifying air) and a central air conditioner (cooling

air).

[0066] In general, dehumidification system **600** attempts to match the saturating temperature of secondary evaporator **640** to the saturating temperature of secondary condenser **620**. The saturating temperature of secondary evaporator **640** and secondary condenser **620** generally is controlled according to the equation: (temperature of inlet air **601**+temperature of second airflow **615**)/2. As the saturating temperature of secondary evaporator **640** is lower than inlet air **601**, evaporation happens in secondary evaporator **640**. As the saturating temperature of secondary condenser **620** is higher than second airflow **615**, condensation happens in secondary condenser **620**. The amount of refrigerant **605** evaporating in secondary evaporator **640** is substantially equal to that condensing in secondary condenser **620**.

[0067] Primary evaporator **610** receives flow of refrigerant **605** from secondary metering device **690** and outputs flow of refrigerant **605** to compressor **660**. Primary evaporator **610** may be any type of coil (e.g., fin tube, micro channel, etc.). Primary evaporator **610** receives first airflow **645** from secondary evaporator **640** and outputs second airflow **615** to secondary condenser **620**. Second airflow **615**, in general, is at a cooler temperature than first airflow **645**. To cool incoming first airflow **645**, primary evaporator **610** transfers heat from first airflow **645** to flow of refrigerant **605**, thereby causing flow of refrigerant **605** to evaporate at least partially from liquid to gas. This transfer of heat from first airflow **645** to flow of refrigerant **605** also removes water from first airflow **645**.

[0068] Secondary condenser **620** receives flow of refrigerant **605** from secondary evaporator **640** and outputs flow of refrigerant **605** to secondary metering device **690**. Secondary condenser **620** may be any type of coil (e.g., fin tube, micro channel, etc.). Secondary condenser **620** receives second airflow **615** from primary evaporator **610** and outputs third airflow **625**. Third airflow **625** is, in general, warmer and drier (i.e., the dew point will be the same but relative humidity will be lower) than second airflow **615**. Secondary condenser **620** generates third airflow **625** by transferring heat from flow of refrigerant **605** to second airflow **615**, thereby causing flow of refrigerant **605** to condense at least partially from gas to liquid. As described above, third airflow **625** is output into the conditioned space. In other embodiments (e.g., as shown in FIGS. **8A** and **8B**), third airflow **625** may first pass through and/or over sub-cooling coil **650** before being output into the conditioned space at a further decreased relative humidity.

[0069] As shown in FIG. **6A**, refrigerant **605** flows outdoors or to an unconditioned space to compressor **660** of condenser system **604**. Alternatively, the refrigerant **605** may continue to flow to the compressor **660** within the dehumidification unit **602** prior to flowing outdoors or to an unconditioned space, as seen in FIG. **6B**. In both FIGS. **6A** and **6B**, compressor **660** pressurizes flow of refrigerant **605**, thereby increasing the temperature of refrigerant **605**. For example, if flow of refrigerant **605** entering compressor **660** is 128 psig/52° F./100% vapor, flow of refrigerant **605** may be 340 psig/150° F./100% vapor as it leaves compressor **660**. Compressor **660** receives flow of refrigerant **605** from primary evaporator **610** and supplies the pressurized flow of refrigerant **605** to primary condenser **630**.

[0070] Primary condenser **630** receives flow of refrigerant **605** from compressor **660** and outputs flow of refrigerant **605** to sub-cooling coil **650**. Primary condenser **630** may be any type of coil (e.g., fin tube, micro channel, etc.). Primary condenser **630** and sub-cooling coil **650** receive first outdoor airflow **606** and output second outdoor airflow **608**. Second outdoor airflow **608** is, in general, warmer (i.e., have a lower relative humidity) than first outdoor airflow **606**. Primary condenser **630** transfers heat from flow of refrigerant **605**, thereby causing flow of refrigerant **605** to condense at least partially from gas to liquid. In some embodiments, primary condenser **630** completely condenses flow of refrigerant **605** to a liquid (i.e., 100% liquid). In other embodiments, primary condenser **630** partially condenses flow of refrigerant **605** to a liquid (i.e., less than 100% liquid).

[0071] Sub-cooling coil **650**, which is an optional component of dehumidification system **600**, sub-

cools the liquid refrigerant **605** as it leaves primary condenser **630**. This, in turn, supplies primary metering device **680** with a liquid refrigerant that is 30 degrees (or more) cooler than before it enters sub-cooling coil **650**. For example, if flow of refrigerant **605** entering sub-cooling coil **650** is 340 psig/105° F./60% vapor, flow of refrigerant **605** may be 340 psig/80° F./0% vapor as it leaves sub-cooling coil **650**. The sub-cooled refrigerant **605** has a greater heat enthalpy factor as well as a greater density, which improves energy transfer between airflow and evaporator resulting in the removal of further latent heat from refrigerant **605**. This further results in greater efficiency and less energy use of dehumidification system **600**. Embodiments of dehumidification system **600** may or may not include a sub-cooling coil **650**.

[0072] In certain embodiments, sub-cooling coil **650** and primary condenser **630** are combined into a single coil. Such a single coil includes appropriate circuiting for flow of airflows **606** and **608** and refrigerant **605**. An illustrative example of a condenser system **604** comprising a single coil condenser and sub-cooling coil is shown in FIG. 7. The single unit coil comprises interior tubes **710** corresponding to the condenser and exterior tubes **705** corresponding to the sub-cooling coil. Refrigerant may be directed through the interior tubes **710** before flowing through exterior tubes **705**. In the illustrative example shown in FIG. 7, airflow is drawn through the single unit coil by fan **695** and expelled upwards. It should be understood, however, that condenser systems of other embodiments can include a condenser, compressor, optional sub-cooling coil, and fan with other configurations known in the art.

[0073] Secondary evaporator **640** receives flow of refrigerant **605** from primary metering device **680** and outputs flow of refrigerant **605** to secondary condenser **620**. Secondary evaporator **640** may be any type of coil (e.g., fin tube, micro channel, etc.). Secondary evaporator **640** receives inlet air **601** and outputs first airflow **645** to primary evaporator **610**. First airflow **645**, in general, is at a cooler temperature than inlet air **601**. To cool incoming inlet air **601**, secondary evaporator **640** transfers heat from inlet air **601** to flow of refrigerant **605**, thereby causing flow of refrigerant **605** to evaporate at least partially from liquid to gas.

[0074] Fan **670** may include any suitable components operable to draw inlet air **601** into dehumidification unit **602** and through secondary evaporator **640**, primary evaporator **610**, and secondary condenser **620**. Fan **670** may be any type of air mover (e.g., axial fan, forward inclined impeller, and backward inclined impeller, etc.). For example, fan **670** may be a backward inclined impeller positioned adjacent to secondary condenser **620**.

[0075] While fan **670** is depicted in FIGS. 6A and 6B as being located adjacent to condenser **620**, it should be understood that fan **670** may be located anywhere along the airflow path of dehumidification unit **602**. For example, fan **670** may be positioned in the airflow path of any one of airflows **601**, **645**, **615**, or **625**. Moreover, dehumidification unit **602** may include one or more additional fans positioned within any one or more of these airflow paths. Similarly, while fan **695** of condenser system **604** is depicted in FIGS. 6A and 6B as being located above primary condenser **630**, it should be understood that fan **695** may be located anywhere (e.g., above, below, beside) with respect to condenser **630** and sub-cooling coil **650**, so long fan **695** is appropriately positioned and configured to facilitate flow of airflow **606** towards primary condenser **630** and sub-cooling coil **650**.

[0076] The rate of airflow generated by fan **670** may be different than that generated by fan **695**. For example, the flow rate of airflow **606** generated by fan **695** may be higher than the flow rate of airflow **601** generated by fan **670**. This difference in flow rates may provide several advantages for the dehumidification systems described herein. For example, a large airflow generated by fan **695** may provide for improved heat transfer at the sub-cooling coil **650** and primary condenser **630** of the condenser system **604**. In general, the rate of airflow generated by second fan **695** is between about 2-times to 5-times that of the rate of airflow generated by first fan **670**. For example, the rate of airflow generated by first fan **670** may be from about 200 to 400 cubic feet per minute (cfm). For example, the rate of airflow generated by second fan **695** may be from about 900 to 1200 cubic feet

per minute (cfm).

[0077] Primary metering device **680** and secondary metering device **690** are any appropriate type of metering/expansion device. In some embodiments, primary metering device **680** is a thermostatic expansion valve (TXV) and secondary metering device **690** is a fixed orifice device (or vice versa). In certain embodiments, metering devices **680** and **690** remove pressure from flow of refrigerant **605** to allow expansion or change of state from a liquid to a vapor in evaporators **610** and **640**. The high-pressure liquid (or mostly liquid) refrigerant entering metering devices **680** and **690** is at a higher temperature than the liquid refrigerant **605** leaving metering devices **680** and **690**. For example, if flow of refrigerant **605** entering primary metering device **680** is 340 psig/80° F./0% vapor, flow of refrigerant **605** may be 196 psig/68° F./5% vapor as it leaves primary metering device **680**. As another example, if flow of refrigerant **605** entering secondary metering device **690** is 196 psig/68° F./4% vapor, flow of refrigerant **605** may be 128 psig/44° F./14% vapor as it leaves secondary metering device **690**.

[0078] In certain embodiments, secondary metering device **690** is operated in a substantially open state (referred to herein as a “fully open” state) such that the pressure of refrigerant **605** entering metering device **690** is substantially the same as the pressure of refrigerant **605** exiting metering device **605**. For example, the pressure of refrigerant **605** may be 80%, 90%, 95%, 99%, or up to 100% of the pressure of refrigerant **605** entering metering device **690**. With the secondary metering device **690** operated in a “fully open” state, primary metering device **680** is the primary source of pressure drop in dehumidification system **600**. In this configuration, airflow **615** is not substantially heated when it passes through secondary condenser **620**, and the secondary evaporator **640**, primary evaporator **610**, and secondary condenser **620** effectively act as a single evaporator. Although, less water may be removed from airflow **601** when the secondary metering device **690** is operated in a “fully open” state, airflow **606** will be output to the conditioned space at a lower temperature than when secondary metering device **690** is not in a “fully open” state. This configuration corresponds to a relatively high sensible heat ratio (SHR) operating mode such that dehumidification system **600** may produce a cool airflow **625** with properties similar to those of an airflow produced by a central air conditioner. If the rate of airflow **601** is increased to a threshold value (e.g., by increasing the speed of fan **670** or one or more other fans of dehumidification system **600**), dehumidification system **600** may perform sensible cooling without removing water from airflow **601**.

[0079] Refrigerant **605** may be any suitable refrigerant such as R410a. In general, dehumidification system **600** utilizes a closed refrigeration loop of refrigerant **605** that passes from compressor **660** through primary condenser **630**, (optionally) sub-cooling coil **650**, primary metering device **680**, secondary evaporator **640**, secondary condenser **620**, secondary metering device **690**, and primary evaporator **610**. Compressor **660** pressurizes flow of refrigerant **605**, thereby increasing the temperature of refrigerant **605**. Primary and secondary condensers **630** and **620**, which may include any suitable heat exchangers, cool the pressurized flow of refrigerant **605** by facilitating heat transfer from the flow of refrigerant **605** to the respective airflows passing through them (i.e., first outdoor airflow **606** and second airflow **615**). The cooled flow of refrigerant **605** leaving primary and secondary condensers **630** and **620** may enter a respective expansion device (i.e., primary metering device **680** and secondary metering device **690**) that is operable to reduce the pressure of flow of refrigerant **605**, thereby reducing the temperature of flow of refrigerant **605**. Primary and secondary evaporators **610** and **640**, which may include any suitable heat exchanger, receive flow of refrigerant **605** from secondary metering device **690** and primary metering device **680**, respectively. Primary and secondary evaporators **610** and **640** facilitate the transfer of heat from the respective airflows passing through them (i.e., inlet air **601** and first airflow **645**) to flow of refrigerant **605**. Flow of refrigerant **605**, after leaving primary evaporator **610**, passes back to compressor **660**, and the cycle is repeated.

[0080] In certain embodiments, the above-described refrigeration loop may be configured such that

evaporators **610** and **640** operate in a flooded state. In other words, flow of refrigerant **605** may enter evaporators **610** and **640** in a liquid state, and a portion of flow of refrigerant **605** may still be in a liquid state as it exits evaporators **610** and **640**. Accordingly, the phase change of flow of refrigerant **605** (liquid to vapor as heat is transferred to flow of refrigerant **605**) occurs across evaporators **610** and **640**, resulting in nearly constant pressure and temperature across the entire evaporators **610** and **640** (and, as a result, increased cooling capacity).

[0081] In operation of example embodiments of dehumidification system **600**, inlet air **601** may be drawn into dehumidification system **600** by fan **670**. Inlet air **601** passes through secondary evaporator **640** in which heat is transferred from inlet air **601** to the cool flow of refrigerant **605** passing through secondary evaporator **640**. As a result, inlet air **601** may be cooled. As an example, if inlet air **601** is 80° F./60% humidity, secondary evaporator **640** may output first airflow **645** at 70° F./84% humidity. This may cause flow of refrigerant **605** to partially vaporize within secondary evaporator **640**. For example, if flow of refrigerant **605** entering secondary evaporator **640** is 196 psig/68° F./5% vapor, flow of refrigerant **605** may be 196 psig/68° F./38% vapor as it leaves secondary evaporator **640**.

[0082] The cooled inlet air **601** leaves secondary evaporator **640** as first airflow **645** and enters primary evaporator **610**. Like secondary evaporator **640**, primary evaporator **610** transfers heat from first airflow **645** to the cool flow of refrigerant **605** passing through primary evaporator **610**. As a result, first airflow **645** may be cooled to or below its dew point temperature, causing moisture in first airflow **645** to condense (thereby reducing the absolute humidity of first airflow **645**). As an example, if first airflow **645** is 70° F./84% humidity, primary evaporator **610** may output second airflow **615** at 54° F./98% humidity. This may cause flow of refrigerant **605** to partially or completely vaporize within primary evaporator **610**. For example, if flow of refrigerant **605** entering primary evaporator **610** is 128 psig/44° F./14% vapor, flow of refrigerant **605** may be 128 psig/52° F./100% vapor as it leaves primary evaporator **610**. In certain embodiments, the liquid condensate from first airflow **645** may be collected in a drain pan connected to a condensate reservoir, as illustrated in FIG. 4. Additionally, the condensate reservoir may include a condensate pump that moves collected condensate, either continually or at periodic intervals, out of dehumidification system **600** (e.g., via a drain hose) to a suitable drainage or storage location.

[0083] The cooled first airflow **645** leaves primary evaporator **610** as second airflow **615** and enters secondary condenser **620**. Secondary condenser **620** facilitates heat transfer from the hot flow of refrigerant **605** passing through the secondary condenser **620** to second airflow **615**. This reheats second airflow **615**, thereby decreasing the relative humidity of second airflow **615**. As an example, if second airflow **615** is 54° F./98% humidity, secondary condenser **620** may output dehumidified airflow **625** at 65° F./68% humidity. This may cause flow of refrigerant **605** to partially or completely condense within secondary condenser **620**. For example, if flow of refrigerant **605** entering secondary condenser **620** is 196 psig/68° F./38% vapor, flow of refrigerant **605** may be 196 psig/68° F./4% vapor as it leaves secondary condenser **620**. In some embodiments, second airflow **615** leaves secondary condenser **620** as dehumidified airflow **625** and is output to a conditioned space.

[0084] Primary condenser **630** facilitates heat transfer from the hot flow of refrigerant **605** passing through the primary condenser **630** to a first outdoor airflow **606**. This heats outdoor airflow **606**, which is output to the unconditioned space (e.g., outdoors) as second outdoor airflow **608**. As an example, if first outdoor airflow **606** is 65° F./68% humidity, primary condenser **630** may output second outdoor airflow **608** at 102° F./19% humidity. This may cause flow of refrigerant **605** to partially or completely condense within primary condenser **630**. For example, if flow of refrigerant **605** entering primary condenser **630** is 340 psig/150° F./100% vapor, flow of refrigerant **605** may be 340 psig/105° F./60% vapor as it leaves primary condenser **630**.

[0085] As described above, some embodiments of dehumidification system **600** may include a sub-cooling coil **650** in the airflow between an inlet of the condenser system **604** and primary

condenser **630**. Sub-cooling coil **650** facilitates heat transfer from the hot flow of refrigerant **605** passing through sub-cooling coil **650** to first outdoor airflow **606**. This heats first outdoor airflow **606**, thereby increasing the temperature of first outdoor airflow **606**. As an example, if first outdoor airflow **606** is 65° F./68% humidity, sub-cooling coil **650** may output an airflow at 81° F./37% humidity. This may cause flow of refrigerant **605** to partially or completely condense within sub-cooling coil **650**. For example, if flow of refrigerant **605** entering sub-cooling coil **650** is 340 psig/150° F./60% vapor, flow of refrigerant **605** may be 340 psig/80° F./0% vapor as it leaves sub-cooling coil **650**.

[0086] In the embodiment depicted in FIGS. **6A** and **6B**, sub-cooling coil **650** is within condenser system **604**. This configuration minimizes the temperature of third airflow **625**, which is output into the conditioned space. An alternative embodiment is shown as dehumidification system **800** of FIGS. **8A** and **8B** in which dehumidification unit **802** includes sub-cooling coil **650**. In these embodiments, airflow **625** first passes through sub-cooling coil **650** before being output to the conditioned space as airflow **855** via fan **670**. As described herein, fan **670** can alternatively be located anywhere along the path of airflow in dehumidification unit **802**, and one or more additional fans can be included in dehumidification unit **802**.

[0087] Without wishing to be bound to any particular theory, the configuration of dehumidification system **800** is believed to be more energy efficient under common operating conditions than that of dehumidification system **600** of FIGS. **6A-6B**. For example, if the temperature of third airflow **625** is less than the outdoor temperature (i.e., the temperature of airflow **606**), then refrigerant **605** will be more effectively cooled, or sub-cooled, with sub-cooling coil **650** placed in the dehumidification unit **802**. Such operating conditions may be common, for example, in locations with warm climates and/or during summer months. As illustrated in FIG. **8B**, indoor dehumidification unit **802** also includes compressor **660**, which may, for example, be located near secondary evaporator **640**, primary evaporator **610**, and/or secondary condenser **620**. In certain embodiments, the dehumidification unit **802** may comprise the compressor **660**, but the dehumidification system **800** may lack the optional sub-cooling coil **650**, as illustrated in FIG. **8C**. The dehumidification system **800** of FIG. **8C** may not require the sub-cooling coil **650** if, for example, the primary condenser **630** is operable to facilitate heat transfer from the flow of refrigerant **605** to a first outdoor airflow **606** in order to effectively condense the refrigerant prior to the flow of refrigerant entering a primary metering device **680**.

[0088] In operation of example embodiments of dehumidification system **800**, as illustrated in each of FIGS. **8A-8C**, inlet air **601** may be drawn into dehumidification system **800** by fan **670**. Inlet air **601** passes through secondary evaporator **640** in which heat is transferred from inlet air **601** to the cool flow of refrigerant **605** passing through secondary evaporator **640**. As a result, inlet air **601** may be cooled. As an example, if inlet air **601** is 80° F./60% humidity, secondary evaporator **640** may output first airflow **645** at 70° F./84% humidity. This may cause flow of refrigerant **605** to partially vaporize within secondary evaporator **640**. For example, if flow of refrigerant **605** entering secondary evaporator **640** is 196 psig/68° F./5% vapor, flow of refrigerant **605** may be 196 psig/68° F./38% vapor as it leaves secondary evaporator **640**.

[0089] The cooled inlet air **601** leaves secondary evaporator **640** as first airflow **645** and enters primary evaporator **610**. Like secondary evaporator **640**, primary evaporator **610** transfers heat from first airflow **645** to the cool flow of refrigerant **605** passing through primary evaporator **610**. As a result, first airflow **645** may be cooled to or below its dew point temperature, causing moisture in first airflow **645** to condense (thereby reducing the absolute humidity of first airflow **645**). As an example, if first airflow **645** is 70° F./84% humidity, primary evaporator **610** may output second airflow **615** at 54° F./98% humidity. This may cause flow of refrigerant **605** to partially or completely vaporize within primary evaporator **610**. For example, if flow of refrigerant **605** entering primary evaporator **610** is 128 psig/44° F./14% vapor, flow of refrigerant **605** may be 128 psig/52° F./100% vapor as it leaves primary evaporator **610**. In certain embodiments, the liquid

condensate from first airflow **645** may be collected in a drain pan connected to a condensate reservoir, as illustrated in FIG. **4**. Additionally, the condensate reservoir may include a condensate pump that moves collected condensate, either continually or at periodic intervals, out of dehumidification system **800** (e.g., via a drain hose) to a suitable drainage or storage location.

[0090] The cooled first airflow **645** leaves primary evaporator **610** as second airflow **615** and enters secondary condenser **620**. Secondary condenser **620** facilitates heat transfer from the hot flow of refrigerant **605** passing through the secondary condenser **620** to second airflow **615**. This reheats second airflow **615**, thereby decreasing the relative humidity of second airflow **615**. As an example, if second airflow **615** is 54° F./98% humidity, secondary condenser **620** may output dehumidified airflow **625** at 65° F./68% humidity. This may cause flow of refrigerant **605** to partially or completely condense within secondary condenser **620**. For example, if flow of refrigerant **605** entering secondary condenser **620** is 196 psig/68° F./38% vapor, flow of refrigerant **605** may be 196 psig/68° F./4% vapor as it leaves secondary condenser **620**. In some embodiments, second airflow **615** leaves secondary condenser **620** as dehumidified airflow **625** and is output to a conditioned space.

[0091] In both FIGS. **8A** and **8B**, dehumidified airflow **625** enters sub-cooling coil **650**, which facilitates heat transfer from the hot flow of refrigerant **605** passing through sub-cooling coil **650** to dehumidified airflow **625**. This heats dehumidified airflow **625**, thereby further decreasing the humidity of dehumidified airflow **625**. As an example, if dehumidified airflow **625** is 65° F./68% humidity, sub-cooling coil **650** may output an airflow **855** at 81° F./37% humidity. This may cause flow of refrigerant **605** to partially or completely condense within sub-cooling coil **650**. For example, if flow of refrigerant **605** entering sub-cooling coil **650** is 340 psig/150° F./60% vapor, flow of refrigerant **605** may be 340 psig/80° F./0% vapor as it leaves sub-cooling coil **650**.

[0092] With reference back to each of FIGS. **8A-8C**, primary condenser **630** facilitates heat transfer from the hot flow of refrigerant **605** passing through the primary condenser **630** to a first outdoor airflow **606**. This heats outdoor airflow **606**, which is output to the unconditioned space as second outdoor airflow **608**. As an example, if first outdoor airflow **606** is 65° F./68% humidity, primary condenser **630** may output second outdoor airflow **608** at 102° F./19% humidity. This may cause flow of refrigerant **605** to partially or completely condense within primary condenser **630**. For example, if flow of refrigerant **605** entering primary condenser **630** is 340 psig/150° F./100% vapor, flow of refrigerant **605** may be 340 psig/105° F./60% vapor as it leaves primary condenser **630**.

[0093] Some embodiments of dehumidification systems **600** and **800** of FIGS. **6A-6B** and **8A-8C** may include a controller that may include one or more computer systems at one or more locations. Each computer system may include any appropriate input devices (such as a keypad, touch screen, mouse, or other device that can accept information), output devices, mass storage media, or other suitable components for receiving, processing, storing, and communicating data. Both the input devices and output devices may include fixed or removable storage media such as a magnetic computer disk, CD-ROM, or other suitable media to both receive input from and provide output to a user. Each computer system may include a personal computer, workstation, network computer, kiosk, wireless data port, personal data assistant (PDA), one or more processors within these or other devices, or any other suitable processing device. In short, the controller may include any suitable combination of software, firmware, and hardware.

[0094] The controller may additionally include one or more processing modules. Each processing module may each include one or more microprocessors, controllers, or any other suitable computing devices or resources and may work, either alone or with other components of dehumidification systems **600** and **800**, to provide a portion or all of the functionality described herein. The controller may additionally include (or be communicatively coupled to via wireless or wireline communication) computer memory. The memory may include any memory or database module and may take the form of volatile or non-volatile memory, including, without limitation,

magnetic media, optical media, random access memory (RAM), read-only memory (ROM), removable media, or any other suitable local or remote memory component.

[0095] Although particular implementations of dehumidification systems **600** and **800** are illustrated and primarily described, the present disclosure contemplates any suitable implementation of dehumidification systems **600** and **800**, according to particular needs. Moreover, although various components of dehumidification systems **600** and **800** have been depicted as being located at particular positions and relative to one another, the present disclosure contemplates those components being positioned at any suitable location, according to particular needs.

[0096] In certain embodiments, the secondary evaporator (**340**, **640**), primary evaporator (**310**, **610**), and secondary condenser (**320**, **620**) of FIG. 3, **6A-6B**, or **8A-8C** are combined in a single coil pack. The single coil pack may include portions (e.g., separate refrigerant circuits) to accommodate the respective functions of secondary evaporator, primary evaporator, and secondary condenser, described above. An illustrative example of such a single coil pack is shown in FIG. 9. FIG. 9 shows a single coil pack **900** which includes a plurality of coils (represented by circles in FIG. 9). Coil pack **900** includes a secondary evaporator portion **940**, primary evaporator portion **910**, and secondary condenser portion **920**. The coil pack may include and/or be fluidly connectable to metering devices **980** and **990** as shown in the exemplary case of FIG. 9. In certain embodiments, metering devices **980** and **990** correspond to primary metering device **380** and secondary metering device **390** of FIG. 3.

[0097] In general, metering devices **980** and **990** may be any appropriate type of metering/expansion device. In some embodiments, metering device **980** is a thermostatic expansion valve (TXV) and secondary metering device **990** is a fixed orifice device (or vice versa). In general, metering devices **980** and **990** remove pressure from flow of refrigerant **905** to allow expansion or change of state from a liquid to a vapor in evaporator portions **910** and **940**. The high-pressure liquid (or mostly liquid) refrigerant **905** entering metering devices **980** and **990** is at a higher temperature than the liquid refrigerant **905** leaving metering devices **980** and **990**. For example, if flow of refrigerant **905** entering metering device **980** is 340 psig/80° F./0% vapor, flow of refrigerant **905** may be 196 psig/68° F./5% vapor as it leaves primary metering device **980**. As another example, if flow of refrigerant **905** entering secondary metering device **990** is 196 psig/68° F./4% vapor, flow of refrigerant **905** may be 128 psig/44° F./14% vapor as it leaves secondary metering device **990**. Refrigerant **905** may be any suitable refrigerant, as described above with respect to refrigerant **305** of FIG. 3.

[0098] In operation of example embodiments of the single coil pack **900**, inlet airflow **901** passes through secondary evaporator portion **940** in which heat is transferred from inlet air **901** to the cool flow of refrigerant **905** passing through secondary evaporator portion **940**. As a result, inlet air **901** may be cooled. As an example, if inlet air **901** is 80° F./60% humidity, secondary evaporator portion **940** may output first airflow at 70° F./84% humidity. This may cause flow of refrigerant **905** to partially vaporize within secondary evaporator portion **940**. For example, if flow of refrigerant **905** entering secondary evaporator portion **940** is 196 psig/68° F./5% vapor, flow of refrigerant **905** may be 196 psig/68° F./38% vapor as it leaves secondary evaporator portion **940**.

[0099] The cooled inlet air **901** proceeds through coil pack **900**, reaching primary evaporator portion **910**. Like secondary evaporator portion **940**, primary evaporator portion **910** transfers heat from airflow **901** to the cool flow of refrigerant **905** passing through primary evaporator portion **910**. As a result, airflow **901** may be cooled to or below its dew point temperature, causing moisture in airflow **901** to condense (thereby reducing the absolute humidity of airflow **901**). As an example, if airflow **901** is 70° F./84% humidity, primary evaporator portion **910** may cool airflow **901** to 54° F./98% humidity. This may cause flow of refrigerant **905** to partially or completely vaporize within primary evaporator portion **910**. For example, if flow of refrigerant **905** entering primary evaporator portion **910** is 128 psig/44° F./14% vapor, flow of refrigerant **905** may be 128 psig/52° F./100% vapor as it leaves primary evaporator portion **910**. In certain embodiments, the

liquid condensate from airflow through primary evaporator portion **910** may be collected in a drain pan connected to a condensate reservoir (e.g., as illustrated in FIG. **4** and described herein). Additionally, the condensate reservoir may include a condensate pump that moves collected condensate, either continually or at periodic intervals, out of coil pack **900** (e.g., via a drain hose) to a suitable drainage or storage location.

[0100] The cooled airflow **901** leaving primary evaporator portion **910** enters secondary condenser portion **920**. Secondary condenser portion **920** facilitates heat transfer from the hot flow of refrigerant **905** passing through the secondary condenser portion **920** to airflow **901**. This reheats airflow **901**, thereby decreasing its relative humidity. As an example, if airflow **901** is 54° F./98% humidity, secondary condenser portion **920** may output an outlet airflow **925** at 65° F./68% humidity. This may cause flow of refrigerant **905** to partially or completely condense within secondary condenser portion **920**. For example, if flow of refrigerant **905** entering secondary condenser portion **920** is 196 psig/68° F./38% vapor, flow of refrigerant **905** may be 196 psig/68° F./4% vapor as it leaves secondary condenser portion **920**. Outlet airflow **925** may, for example, enter primary condenser portion **330** or sub-cooling coil **350** of FIG. **3**.

[0101] Although a particular implementation of coil pack **900** is illustrated and primarily described, the present disclosure contemplates any suitable implementation of coil pack **900**, according to particular needs. Moreover, although various components of coil pack **900** have been depicted as being located at particular positions, the present disclosure contemplates those components being positioned at any suitable location, according to particular needs.

[0102] In certain embodiments, secondary evaporator (**340**, **640**) and secondary condenser (**320**, **620**) of FIG. **3**, **6A-6B**, or **8A-8C** are combined in a single coil pack such that the single coil pack includes portions (e.g., separate refrigerant circuits) to accommodate the respective functions of the secondary evaporator and secondary condenser. An illustrative example of such an embodiment is shown in FIG. **10**. FIG. **10** shows a single coil pack **1000** which includes a secondary evaporator portion **1040** and secondary condenser portion **1020**. As shown in the illustrative example of FIG. **10**, a primary evaporator **1010** is located between the secondary evaporator portion **1040** and secondary condenser portion **1020** of the single coil pack **1000**. In this exemplary embodiment, the single coil pack **1000** is shown as a “U”-shaped coil. However, alternate embodiments may be used as long as flow airflow **1001** passes sequentially through secondary evaporator portion **1040**, primary evaporator **1010**, and secondary condenser portion **1020**. In general, single coil pack **1000** can include the same or a different coil type compared to that of primary evaporator **1010**. For example, single coil pack **1000** may include a microchannel coil type, while primary evaporator **1010** may include a fin tube coil type. This may provide further flexibility for optimizing a dehumidification system in which single coil pack **1000** and primary evaporator **1010** are used.

[0103] In operation of example embodiments of the single coil pack **1000**, inlet air **1001** passes through secondary evaporator portion **1040** in which heat is transferred from inlet air **1001** to the cool flow of refrigerant passing through secondary evaporator portion **1040**. As a result, inlet air **1001** may be cooled. As an example, if inlet air **1001** is 80° F./60% humidity, secondary evaporator portion **1040** may output airflow at 70° F./84% humidity. This may cause flow of refrigerant to partially vaporize within secondary evaporator portion **1040**. For example, if flow of refrigerant entering secondary evaporator **1040** is 196 psig/68° F./5% vapor, flow of refrigerant **1005** may be 196 psig/68° F./38% vapor as it leaves secondary evaporator portion **1040**.

[0104] The cooled inlet air **1001** leaves secondary evaporator portion **1040** and enters primary evaporator **1010**. Like secondary evaporator portion **1040**, primary evaporator **1010** transfers heat from airflow **1001** to the cool flow of refrigerant passing through primary evaporator **1010**. As a result, airflow **1001** may be cooled to or below its dew point temperature, causing moisture in airflow **1001** to condense (thereby reducing the absolute humidity of airflow **1001**). As an example, if airflow **1001** entering primary evaporator **1010** is 70° F./84% humidity, primary evaporator **1010** may output airflow at 54° F./98% humidity. This may cause flow of refrigerant to partially or

completely vaporize within primary evaporator **1010**. For example, if flow of refrigerant entering primary evaporator **1010** is 128 psig/44° F./14% vapor, flow of refrigerant may be 128 psig/52° F./100% vapor as it leaves primary evaporator **1010**. In certain embodiments, the liquid condensate from airflow **1010** may be collected in a drain pan connected to a condensate reservoir, as illustrated in FIG. 4. Additionally, the condensate reservoir may include a condensate pump that moves collected condensate, either continually or at periodic intervals, out of primary evaporator **1010**, and the associated dehumidification system (e.g., via a drain hose) to a suitable drainage or storage location.

[0105] The cooled airflow **1001** leaves primary evaporator **1010** and enters secondary condenser portion **1020**. Secondary condenser portion **1020** facilitates heat transfer from the hot flow of refrigerant passing through the secondary condenser **1020** to airflow **1001**. This reheats airflow **1001**, thereby decreasing its relative humidity. As an example, if airflow **1001** entering secondary condenser portion **1020** is 54° F./98% humidity, secondary condenser **1020** may output airflow **1025** at 65° F./68% humidity. This may cause flow of refrigerant to partially or completely condense within secondary condenser **1020**. For example, if flow of refrigerant entering secondary condenser portion **1020** is 196 psig/68° F./38% vapor, flow of refrigerant may be 196 psig/68° F./4% vapor as it leaves secondary condenser **1020**. Outlet airflow **925** may, for example, enter primary condenser **330** or sub-cooling cooling **350** of FIG. 3.

[0106] Although a particular implementation of coil pack **1000** is illustrated and primarily described, the present disclosure contemplates any suitable implementation of coil pack **1000**, according to particular needs. Moreover, although various components of coil pack **1000** have been depicted as being located at particular positions, the present disclosure contemplates those components being positioned at any suitable location, according to particular needs.

[0107] In certain embodiments, one or both of the secondary evaporator (**340**, **640**) and primary evaporator (**310**, **610**) of FIG. 3, **6A-6B**, or **8A-8C** are subdivided into two or more circuits. In such embodiments, each circuit of the subdivided evaporator(s) is fed refrigerant by a corresponding metering device. The metering devices may include passive metering devices, active metering devices, or combinations thereof. For example, metering device **380** (or **690**) may be an active thermostatic expansion valve (TXV) and secondary metering device **390** (or **690**) may be a passive fixed orifice device (or vice versa). The metering devices may be configured to feed refrigerant to each circuit within the evaporators at a desired mass flow rate. Metering devices for feeding refrigerant to each circuit of the subdivided evaporator(s) may be used in combination with metering devices **380** and **390** or may replace one or both of metering devices **380** and **390**.

[0108] FIGS. **11**, **12**, **13**, and **14** show an illustrative example of a portion **1100** of a dehumidification system in which the primary evaporator **1110** comprises three circuits for flow of refrigerant, according to certain embodiments. Portion **1100** includes a primary metering device **1180**, secondary metering devices **1190a-c**, a secondary evaporator **1140**, a primary evaporator **1110**, and a secondary condenser **1120**. Primary evaporator **1110** includes three circuits for receiving flow of refrigerant from secondary metering devices **1190a-c**. In the example of FIGS. **11**, **12**, **13**, and **14**, each of secondary metering devices **1190a-c** is a passive metering device (i.e., with an orifice of a fixed inner diameter and length). It should, however be understood that one or more (up to all) of the secondary metering devices **1190a-c** may be active metering devices (e.g., thermostatic expansion valves).

[0109] In operation of example embodiments of portion **1100** of a dehumidification system, flow of cooled (or sub-cooled) refrigerant is received at inlet **1102**, for example, from sub-cooling coil **350** or primary condenser **330** of dehumidification system **300** of FIG. 3. Primary metering device **1180** determines the flow rate of refrigerant into secondary evaporator **1140**. While FIGS. **11**, **12**, **13**, and **14** are shown to have a single primary metering device **1180**, other embodiments can include multiple primary metering devices in parallel (e.g., if the secondary evaporator **1140** comprises two or more circuits for flow of refrigerant).

[0110] As the cooled refrigerant passes through secondary evaporator **1140**, heat is exchanged between the refrigerant and airflow passing through secondary evaporator **1140**, cooling the inlet air. As an example, if inlet air is 80° F./60% humidity, secondary evaporator **1140** may output airflow at 70° F./84% humidity. This may cause flow of refrigerant to partially vaporize within secondary evaporator **1140**. For example, if flow of refrigerant entering secondary evaporator **1140** is 196 psig/68° F./5% vapor, flow of refrigerant may be 196 psig/68° F./38% vapor as it leaves secondary evaporator **1140**.

[0111] Secondary condenser **1120** receives warmed refrigerant from secondary evaporator **1140** via tube **1106**. Secondary condenser **1120** facilitates heat transfer from the hot flow of refrigerant passing through the secondary condenser **1120** to the airflow. This reheats the airflow, thereby decreasing its relative humidity. As an example, if the airflow is 54° F./98% humidity, secondary condenser **1120** may output an airflow at 65° F./68% humidity. This may cause flow of refrigerant to partially or completely condense within secondary condenser **1120**. For example, if flow of refrigerant entering secondary condenser **1120** is 196 psig/68° F./38% vapor, flow of refrigerant may be 196 psig/68° F./4% vapor as it leaves secondary condenser **1120**.

[0112] The cooled refrigerant exits the secondary condenser at **1108** and is received by metering devices **1190a-c**, which distributes the flow of refrigerant into the three circuits of primary evaporator **1110**. FIG. **14** shows a view which includes the circuiting of primary evaporator **1110**. Airflow passing through primary evaporator **1110** may be cooled to or below its dew point temperature, causing moisture in the airflow to condense (thereby reducing the absolute humidity of the air). As an example, if the airflow is 70° F./84% humidity, primary evaporator **1110** may output airflow at 54° F./98% humidity. This may cause flow of refrigerant to partially or completely vaporize within primary evaporator **1110**.

[0113] Each of secondary metering devices **1190a**, **1190b**, and **1190c** is configured to provide flow of refrigerant to each circuit of primary evaporator **1110** at a desired flow rate. For example, the flow rate provided to each circuit may be optimized to improve performance of the primary evaporator **1110**. For example, under certain operating conditions, it may be beneficial to prevent the entire flow of refrigerant from passing through the entire evaporator, as occurs in a traditional evaporator coil. Refrigerant flowing through such an evaporator might undergo a change from liquid to gas phase before exiting the coil, resulting in poor performance in the portion of the evaporator that only contacts gaseous refrigerant. To significantly reduce or eliminate this problem, the present disclosure provides for refrigerant flow at a desired flow rate through each circuit. The desired flow rate may be predetermined (e.g., based on known design criteria and/or operating conditions) and/or variable (e.g., manually and/or automatically adjustable in real time) during operation. The flow rate may be configured such that the flow of refrigerant exits its respective circuit just after transitioning to a gas. For example, the rate of airflow near the edges of an evaporator may be less than near the center of the evaporator. Therefore, a lower rate of refrigerant flow may be supplied by secondary metering devices **1190a-c** to the circuits corresponding to the edge of primary evaporator **1110**.

[0114] While the example of FIGS. **11**, **12**, **13**, and **14** include a primary evaporator that is subdivided into two or more circuits. In other embodiments, secondary evaporator **1110** may also, or alternatively, be subdivided into two or more circuits. It should also be appreciated that the circuiting exemplified by FIGS. **11**, **12**, **13**, and **14** can also be achieved in single coil packs such as those shown in FIGS. **9** and **10**.

[0115] Although a particular implementation of portion **1100** of a dehumidification system is illustrated and primarily described, the present disclosure contemplates any suitable implementation of portion **1100** of a dehumidification system, according to particular needs. Moreover, although various components of portion **1100** of a dehumidification system have been depicted as being located at particular positions, the present disclosure contemplates those components being positioned at any suitable location, according to particular needs.

[0116] FIGS. 15A-15B illustrate an example dehumidification system **1500** that may be used in accordance with dehumidification system **300** of FIG. 3 to reduce the humidity of air within a structure. Dehumidification system **1500** includes a dehumidification unit **1502**, which is generally indoors, and a heat exchanger **1504** or an external source **1506** configured to contain a volume of a fluid operable to be used by the dehumidification system **1500** to cool a separate fluid flow within the dehumidification unit **1502**. FIG. 15A illustrates the dehumidification system **1500** comprising the heat exchanger **1504**, and FIG. 15B illustrates the dehumidification system comprising the external source **1506**. With reference to both FIGS. 15A-15B, dehumidification unit **1502** includes a primary evaporator **1508**, a primary condenser **1510**, a secondary evaporator **1512**, a secondary condenser **1514**, a compressor **1516**, a primary metering device **1518**, a secondary metering device **1520**, and a fan **1522**.

[0117] With continued reference to both FIGS. 15A-15B, a flow of refrigerant **1524** is circulated through dehumidification unit **1502** as illustrated. In general, dehumidification unit **1502** receives an inlet airflow **1526**, removes water from inlet airflow **1526**, and discharges dehumidified air **1528**. Water is removed from inlet air **1526** using a refrigeration cycle of flow of refrigerant **1524**. By including secondary evaporator **1512** and secondary condenser **1514**, however, dehumidification system **1500** causes at least part of the flow of refrigerant **1524** to evaporate and condense twice in a single refrigeration cycle. This increases the refrigeration capacity over typical systems without adding any additional power to the compressor, thereby increasing the overall dehumidification efficiency of the system.

[0118] In general, dehumidification system **1500** attempts to match the saturating temperature of secondary evaporator **1512** to the saturating temperature of secondary condenser **1514**. The saturating temperature of secondary evaporator **1512** and secondary condenser **1514** generally is controlled according to the equation: $(\text{temperature of inlet air } 1526 + \text{temperature of a second airflow } 1530) / 2$. As the saturating temperature of secondary evaporator **1512** is lower than inlet air **1526**, evaporation happens in secondary evaporator **1512**. As the saturating temperature of secondary condenser **1514** is higher than second airflow **1530**, condensation happens in the secondary condenser **1514**. The amount of refrigerant **1524** evaporating in secondary evaporator **1512** is substantially equal to that condensing in secondary condenser **1514**.

[0119] Primary evaporator **1508** receives flow of refrigerant **1524** from secondary metering device **1520** and outputs flow of refrigerant **1524** to compressor **1516**. Primary evaporator **1508** may be any suitable type of coil (e.g., fin tube, micro channel, etc.). Primary evaporator **1508** receives a first airflow **1532** from secondary evaporator **1512** and outputs second airflow **1530** to secondary condenser **1514**. Second airflow **1530**, in general, is at a cooler temperature than first airflow **1532**. To cool incoming first airflow **1532**, primary evaporator **1508** transfers heat from first airflow **1532** to flow of refrigerant **1524**, thereby causing flow of refrigerant **1524** to evaporate at least partially from liquid to gas. This transfer of heat from first airflow **1532** to flow of refrigerant **1524** also removes water from first airflow **1532**.

[0120] Secondary condenser **1514** receives flow of refrigerant **1524** from secondary evaporator **1512** and outputs flow of refrigerant **1524** to secondary metering device **1520**. Secondary condenser **1514** may be any type of coil (e.g., fin tube, micro channel, etc.). Secondary condenser **1514** receives second airflow **1530** from primary evaporator **1508** and outputs dehumidified airflow **1528**. Dehumidified airflow **1528** is, in general, warmer and drier (i.e., the dew point will be the same but relative humidity will be lower) than second airflow **1530**. Secondary condenser **1514** generates dehumidified airflow **1528** by transferring heat from flow of refrigerant **1524** to second airflow **1530**, thereby causing flow of refrigerant **1524** to condense at least partially from gas to liquid.

[0121] Primary condenser **1510** receives flow of refrigerant **1524** from compressor **1516** and outputs flow of refrigerant **1524** to primary metering device **1518**. Primary condenser **1510** may be any type of liquid-cooled heat exchanger operable to transfer heat from the flow of refrigerant **1524**

to the flow of a fluid **1534**. In embodiments, the fluid **1534** may be any suitable fluid, such as water or a mixture of water and glycol. Primary condenser **1510** receives both the flow of fluid **1534** and the flow of refrigerant **1524** during operation of dehumidification system **1500**, wherein the primary condenser **1510** is operable to transfer heat from the flow of refrigerant **1524**, thereby causing flow of refrigerant **1524** to condense at least partially from gas to liquid. In some embodiments, primary condenser **1510** completely condenses flow of refrigerant **1524** to a liquid (i.e., 100% liquid). In other embodiments, primary condenser **1510** partially condenses flow of refrigerant **1524** to a liquid (i.e., less than 100% liquid).

[0122] As illustrated, the dehumidification system **1500** may further comprise a first water pump **1536**. The first water pump **1536** may be disposed internal or external to the dehumidification unit **1502**. The first water pump **1536** may be any suitable device operable to provide for the flow of fluid **1534**. As depicted in FIG. 15A, the first water pump **1536** may be disposed at any suitable position in relation to the primary condenser **1510** and the heat exchanger **1504** operable to cycle the flow of fluid **1534** between the heat exchanger **1504** and the primary condenser **1510**. As depicted in FIG. 15B, the first water pump **1536** may be disposed at any suitable position in relation to the primary condenser **1510** and the external source **1506** operable to cycle the flow of fluid **1534** between the external source **1506** and the primary condenser **1510**.

[0123] With reference to FIG. 15A, heat exchanger **1504** may receive the flow of fluid **1534** from primary condenser **1510** at a first temperature and output flow of fluid **1534** to primary condenser **1510** at a second temperature after transferring heat away from the flow of fluid **1534**, wherein the second temperature is lower than the first temperature. Heat exchanger **1504** may be any suitable type of heat exchanger, such as, for example, a cooling tower or a dry cooler. Heat exchanger **1504** receives the flow of fluid **1534** and a first outdoor airflow **1540**, wherein heat is transferred between the flow of fluid **1534** and the first outdoor airflow **1540**. Heat exchanger **1504** may further output the flow of fluid **1534** and a second outdoor airflow **1542**, wherein the flow of fluid **1534** leaving the heat exchanger **1504** is at a lower temperature than the flow of fluid **1534** received by the heat exchanger **1504**, and the second outdoor airflow **1542** is at a greater temperature than the first outdoor airflow **1540**.

[0124] In embodiments wherein the heat exchanger **1504** is a cooling tower, the heat exchanger **1504** may be operable to dispense the flow of fluid **1534** within its internal structure, wherein the fluid **1534** directly contacts the first outdoor airflow **1540** as the fluid **1534** flows through the heat exchanger **1504** and transfers heat to the first outdoor airflow **1540**. At least a portion of the fluid **1534** may evaporate and exit to the atmosphere as the heat transfers from the fluid **1534** to the first outdoor airflow **1540**, and the heat exchanger **1504** may collect a remaining portion of the fluid **1534** after transferring heat to the first outdoor airflow **1540**, wherein the remaining portion of the fluid **1534** is at a lower temperature. In embodiments wherein the heat exchanger **1504** is a dry cooler, the heat exchanger **1504** may be operable to induce the first outdoor airflow **1540** to flow through the heat exchanger **1504** where heat transfers indirectly between the first outdoor airflow **1540** and the flow of fluid **1534**. In these embodiments, heat transfer would not result in loss of a portion of the fluid **1534** through evaporation to the atmosphere.

[0125] With reference now to FIG. 15B, external source **1506** may receive the flow of fluid **1534** from the primary condenser **1510** and output flow of fluid **1534** to the primary condenser **1510** via first water pump **1536**. External source **1506** may be configured to contain and/or store a volume of fluid **1534** to be used by primary condenser **1510** to lower the temperature of the flow of refrigerant **1524** in the dehumidification unit **1502**. The external source **1506** may be configured to receive the flow of fluid **1534** from primary condenser **1510** at a first temperature and output flow of fluid **1534** to primary condenser **1510** at a second temperature after transferring heat away from the flow of fluid **1534**, wherein the second temperature is lower than the first temperature. Without limitations, the external source **1506** may be any suitable number and combination of a ground reservoir, a natatorium, and an outdoor body of water, among others. In embodiments wherein the

external source **1506** is a ground reservoir, the external source **1506** may implement an open or closed ground water system, wherein the conduit providing for the flow of fluid **1534** within the ground reservoir may be disposed substantially parallel to a horizontal plane of the ground surface, substantially perpendicular to the horizontal plane of the ground surface, or combinations thereof. [0126] With reference to both FIGS. **15A-15B**, secondary evaporator **1512** receives flow of refrigerant **1524** from primary metering device **1518** and outputs flow of refrigerant **1524** to secondary condenser **1514**. Secondary evaporator **1512** may be any type of coil (e.g., fin tube, micro channel, etc.). Secondary evaporator **1512** receives inlet air **1526** and outputs first airflow **1532** to primary evaporator **1508**. First airflow **1532**, in general, is at a cooler temperature than inlet air **1526**. To cool incoming inlet air **1526**, secondary evaporator **1512** transfers heat from inlet air **1526** to flow of refrigerant **1524**, thereby causing flow of refrigerant **1524** to evaporate at least partially from liquid to gas.

[0127] Compressor **1516** pressurizes flow of refrigerant **1524**, thereby increasing the temperature of refrigerant **1524**. For example, if flow of refrigerant **1524** entering compressor **1516** is 128 psig/52° F./100% vapor, flow of refrigerant **1524** may be 340 psig/150° F./100% vapor as it leaves compressor **1516**. Compressor **1516** receives flow of refrigerant **1524** from primary evaporator **1508** and supplies the pressurized flow of refrigerant **1524** to primary condenser **1510**.

[0128] Fan **1522** may include any suitable components operable to draw inlet air **1526** into dehumidification unit **1502** and through secondary evaporator **1512**, primary evaporator **1508**, and secondary condenser **1514**. Fan **1522** may be any type of air mover (e.g., axial fan, forward inclined impeller, and backward inclined impeller, etc.). For example, fan **1522** may be a backward inclined impeller positioned adjacent to secondary condenser **1514**. While fan **1522** is depicted as being located adjacent to secondary condenser **1514**, it should be understood that fan **1522** may be located anywhere along the airflow path of dehumidification unit **1502**. For example, fan **1522** may be positioned in the airflow path of any one of airflows **1526**, **1532**, **1530**, or **1528**. Moreover, dehumidification unit **1502** may include one or more additional fans positioned within any one or more of these airflow paths.

[0129] Primary metering device **1518** and secondary metering device **1520** are any appropriate type of metering/expansion device. In some embodiments, primary metering device **1518** is a thermostatic expansion valve (TXV) and secondary metering device **1520** is a fixed orifice device (or vice versa). In certain embodiments, metering devices **1518** and **1520** remove pressure from flow of refrigerant **1524** to allow expansion or change of state from a liquid to a vapor in evaporators **1512** and **1508**. The high-pressure liquid (or mostly liquid) refrigerant **1524** entering metering devices **1518** and **1520** is at a higher temperature than the liquid refrigerant **1524** leaving metering devices **1518** and **1520**. For example, if flow of refrigerant **1524** entering primary metering device **1518** is 340 psig/80° F./0% vapor, flow of refrigerant **1524** may be 196 psig/68° F./5% vapor as it leaves primary metering device **1518**. As another example, if flow of refrigerant **1524** entering secondary metering device **1520** is 196 psig/68° F./4% vapor, flow of refrigerant **1524** may be 128 psig/44° F./14% vapor as it leaves secondary metering device **1520**.

[0130] Refrigerant **1524** may be any suitable refrigerant such as R410a. In general, dehumidification system **1500** utilizes a closed refrigeration loop of refrigerant **1524** that passes from compressor **1516** through primary condenser **1510**, primary metering device **1518**, secondary evaporator **1512**, secondary condenser **1514**, secondary metering device **1520**, and primary evaporator **1508**. Compressor **1516** pressurizes flow of refrigerant **1524**, thereby increasing the temperature of refrigerant **1524**. Primary condenser **1510**, which may include any suitable water-cooled heat exchanger, cools the pressurized flow of refrigerant **1524** by facilitating heat transfer from the flow of refrigerant **1524** to the flow of fluid provided by the external source **1506** passing through it (i.e., flow of fluid **1534**). Secondary condenser, which may include any suitable air-cooled heat exchanger, cools the pressurized flow of refrigerant **1524** by facilitating heat transfer from the flow of refrigerant **1524** to the respective airflow passing through it (i.e., second airflow

1530).

[0131] The cooled flow of refrigerant **1524** leaving primary and secondary condensers **1510** and **1514** may enter a respective expansion device (i.e., primary metering device **1518** and secondary metering device **1520**) that is operable to reduce the pressure of flow of refrigerant **1524**, thereby reducing the temperature of flow of refrigerant **1524**. Primary and secondary evaporators **1508** and **1512**, which may include any suitable heat exchanger, receive flow of refrigerant **1524** from secondary metering device **1520** and primary metering device **1518**, respectively. Primary and secondary evaporators **1508** and **1512** facilitate the transfer of heat from the respective airflows passing through them (i.e., inlet air **1526** and first airflow **1532**) to flow of refrigerant **1524**. Flow of refrigerant **1524**, after leaving primary evaporator **1508**, passes back to compressor **1516**, and the cycle is repeated.

[0132] In certain embodiments, the above-described refrigeration loop may be configured such that evaporators **1508** and **1512** operate in a flooded state. In other words, flow of refrigerant **1524** may enter evaporators **1508** and **1512** in a liquid state, and a portion of flow of refrigerant **1524** may still be in a liquid state as it exits evaporators **1508** and **1512**. Accordingly, the phase change of flow of refrigerant **1524** (liquid to vapor as heat is transferred to flow of refrigerant **1524**) occurs across evaporators **1508** and **1512**, resulting in nearly constant pressure and temperature across the entire evaporators **1508** and **1512** (and, as a result, increased cooling capacity).

[0133] In operation of example embodiments of dehumidification system **1500**, inlet air **1526** may be drawn into dehumidification unit **1502** by fan **1522**. Inlet air **1526** passes through secondary evaporator **1512** in which heat is transferred from inlet air **1526** to the cool flow of refrigerant **1524** passing through secondary evaporator **1512**. As a result, inlet air **1526** may be cooled. As an example, if inlet air **1526** is 80° F./60% humidity, secondary evaporator **1512** may output first airflow **1532** at 70° F./84% humidity. This may cause flow of refrigerant **1524** to partially vaporize within secondary evaporator **1512**. For example, if flow of refrigerant **1524** entering secondary evaporator **1512** is 196 psig/68° F./5% vapor, flow of refrigerant **1524** may be 196 psig/68° F./38% vapor as it leaves secondary evaporator **1512**.

[0134] The cooled inlet air **1526** leaves secondary evaporator **1512** as first airflow **1532** and enters primary evaporator **1508**. Like secondary evaporator **1512**, primary evaporator **1508** transfers heat from first airflow **1532** to the cool flow of refrigerant **1524** passing through primary evaporator **1508**. As a result, first airflow **1532** may be cooled to or below its dew point temperature, causing moisture in first airflow **1532** to condense (thereby reducing the absolute humidity of first airflow **1532**). As an example, if first airflow **1532** is 70° F./84% humidity, primary evaporator **1508** may output second airflow **1530** at 54° F./98% humidity. This may cause flow of refrigerant **1524** to partially or completely vaporize within primary evaporator **1508**. For example, if flow of refrigerant **1524** entering primary evaporator **1508** is 128 psig/44° F./14% vapor, flow of refrigerant **1524** may be 128 psig/52° F./100% vapor as it leaves primary evaporator **1508**.

[0135] The cooled first airflow **1532** leaves primary evaporator **1508** as second airflow **1530** and enters secondary condenser **1514**. Secondary condenser **1514** facilitates heat transfer from the hot flow of refrigerant **1524** passing through the secondary condenser **1514** to second airflow **1530**. This reheats second airflow **1530**, thereby decreasing the relative humidity of second airflow **1530**. As an example, if second airflow **1530** is 54° F./98% humidity, secondary condenser **1514** may output dehumidified airflow **1528** at 65° F./68% humidity. This may cause flow of refrigerant **1524** to partially or completely condense within secondary condenser **1514**. For example, if flow of refrigerant **1524** entering secondary condenser **1514** is 196 psig/68° F./38% vapor, flow of refrigerant **1524** may be 196 psig/68° F./4% vapor as it leaves secondary condenser **1514**.

[0136] Some embodiments of dehumidification system **1500** may include a controller that may include one or more computer systems at one or more locations. Each computer system may include any appropriate input devices (such as a keypad, touch screen, mouse, or other device that can accept information), output devices, mass storage media, or other suitable components for

receiving, processing, storing, and communicating data. Both the input devices and output devices may include fixed or removable storage media such as a magnetic computer disk, CD-ROM, or other suitable media to both receive input from and provide output to a user. Each computer system may include a personal computer, workstation, network computer, kiosk, wireless data port, personal data assistant (PDA), one or more processors within these or other devices, or any other suitable processing device. In short, the controller may include any suitable combination of software, firmware, and hardware.

[0137] The controller may additionally include one or more processing modules. Each processing module may each include one or more microprocessors, controllers, or any other suitable computing devices or resources and may work, either alone or with other components of dehumidification system **1500**, to provide a portion or all of the functionality described herein. The controller may additionally include (or be communicatively coupled to via wireless or wireline communication) computer memory. The memory may include any memory or database module and may take the form of volatile or non-volatile memory, including, without limitation, magnetic media, optical media, random access memory (RAM), read-only memory (ROM), removable media, or any other suitable local or remote memory component.

[0138] Although particular implementations of dehumidification system **1500** are illustrated and primarily described, the present disclosure contemplates any suitable implementation of dehumidification system **1500**, according to particular needs. Moreover, although various components of dehumidification system **1500** have been depicted as being located at particular positions and relative to one another, the present disclosure contemplates those components being positioned at any suitable location, according to particular needs.

[0139] FIGS. **16A**, **16B**, **16C**, and **16D** illustrate an example dehumidification system **1600** with a modulating valve **1602** that may be used in accordance with split dehumidification system **600** of FIGS. **6A-6B** to reduce humidity of an airflow. Dehumidification system **1600** includes the modulating valve **1602**, a primary evaporator **1604**, a primary condenser **1606**, a secondary evaporator **1608**, a secondary condenser **1610**, a compressor **1612**, a primary metering device **1614**, a secondary metering device **1616**, a fan **1618**, and an alternate condenser **1620**. In some embodiments, dehumidification system **1600** may additionally include an optional sub-cooling coil **1622**. As illustrated in FIGS. **16A-16B**, the alternate condenser **1620** may be disposed in an external condenser unit **1624**. With reference to FIG. **16A**, the optional sub-cooling coil **1622** may be disposed in the external condenser unit **1624** with the alternate condenser **1620**, wherein the sub-cooling coil **1622** and the alternate condenser **1620** may be combined into a single coil. With reference to FIG. **16B**, the optional sub-cooling coil **1622** may be disposed adjacent to the primary condenser **1606**, wherein sub-cooling coil **1620** and primary condenser **1606** may be combined into a single coil. FIGS. **16C-16D** illustrate an embodiment of dehumidification system **1600** wherein both optional sub-cooling coil **1622** and alternate condenser **1620** are not in the external condenser unit **1624** and where alternate condenser **1620** is liquid-cooled.

[0140] With reference to each of FIGS. **16A-16D**, a flow of refrigerant **1626** is circulated through dehumidification system **1600** as illustrated. In general, dehumidification system **1600** receives inlet airflow **1628**, removes water from inlet airflow **1628**, and discharges dehumidified air **1630**. Water is removed from inlet air **1628** using a refrigeration cycle of flow of refrigerant **1626**. By including secondary evaporator **1608** and secondary condenser **1610**, however, dehumidification system **1600** causes at least part of the flow of refrigerant **1626** to evaporate and condense twice in a single refrigeration cycle. This increases the refrigeration capacity over typical systems without adding any additional power to the compressor, thereby increasing the overall dehumidification efficiency of the system.

[0141] In general, dehumidification system **1600** attempts to match the saturating temperature of secondary evaporator **1608** to the saturating temperature of secondary condenser **1610**. The saturating temperature of secondary evaporator **1608** and secondary condenser **1610** generally is

controlled according to the equation: (temperature of inlet air **1628**+temperature of a second airflow **1632**)/2. As the saturating temperature of secondary evaporator **1608** is lower than inlet air **1628**, evaporation happens in secondary evaporator **1608**. As the saturating temperature of secondary condenser **1610** is higher than second airflow **1632**, condensation happens in the secondary condenser **1610**. The amount of refrigerant **1626** evaporating in secondary evaporator **1608** is substantially equal to that condensing in secondary condenser **1610**.

[0142] Primary evaporator **1604** receives flow of refrigerant **1626** from secondary metering device **1616** and outputs flow of refrigerant **1626** to compressor **1612**. Primary evaporator **1604** may be any type of coil (e.g., fin tube, micro channel, etc.). Primary evaporator **1604** receives a first airflow **1634** from secondary evaporator **1608** and outputs second airflow **1632** to secondary condenser **1610**. Second airflow **1632**, in general, is at a cooler temperature than first airflow **1634**. To cool incoming first airflow **1634**, primary evaporator **1604** transfers heat from first airflow **1634** to flow of refrigerant **1626**, thereby causing flow of refrigerant **1626** to evaporate at least partially from liquid to gas. This transfer of heat from first airflow **1634** to flow of refrigerant **1626** also removes water from first airflow **1634**.

[0143] Secondary condenser **1610** receives flow of refrigerant **1626** from secondary evaporator **1608** and outputs flow of refrigerant **1626** to secondary metering device **1616**. Secondary condenser **1610** may be any type of coil (e.g., fin tube, micro channel, etc.). Secondary condenser **1610** receives second airflow **1632** from primary evaporator **1604** and outputs a third airflow **1636**. Third airflow **1636** is, in general, warmer and drier (i.e., the dew point will be the same but relative humidity will be lower) than second airflow **1632**. Secondary condenser **1610** generates third airflow **1632** by transferring heat from flow of refrigerant **1626** to second airflow **1632**, thereby causing flow of refrigerant **1626** to condense at least partially from gas to liquid.

[0144] Primary condenser **1606** may be any type of coil (e.g., fin tube, micro channel, etc.). Primary condenser **1606** is operable to receive flow of refrigerant **1626** from modulating valve **1602** and outputs flow of refrigerant **1626** to either primary metering device **1614** or sub-cooling coil **1622**. As shown in FIG. **16A**, primary condenser **1606** outputs flow of refrigerant **1626** to primary metering device **1614**. In these embodiments, primary condenser **1606** receives third airflow **1636** and outputs dehumidified air **1630**. But with reference to FIGS. **16B-16D**, primary condenser **1606** outputs flow of refrigerant **1626** to the optional sub-cooling coil **1622** before the flow of refrigerant **1626** flows to primary metering device **1614**. In these embodiments, primary condenser **1606** receives a fourth airflow **1638** generated by the sub-cooling coil **1622** and outputs dehumidified air **1630**. With reference to each of FIGS. **16A-16D**, dehumidified air **1630** is, in general, warmer and drier (i.e., have a lower relative humidity) than either third airflow **1636** or fourth airflow **1638**. Primary condenser **1606** generates dehumidified air **1630** by transferring heat away from flow of refrigerant **1626**, thereby causing flow of refrigerant **1626** to condense at least partially from gas to liquid. In some embodiments, primary condenser **1606** completely condenses flow of refrigerant **1626** to a liquid (i.e., 100% liquid). In other embodiments, primary condenser **1606** partially condenses flow of refrigerant **1626** to a liquid (i.e., less than 100% liquid).

[0145] Secondary evaporator **1608** receives flow of refrigerant **1626** from primary metering device **1614** and outputs flow of refrigerant **1626** to secondary condenser **1610**. Secondary evaporator **1608** may be any type of coil (e.g., fin tube, micro channel, etc.). Secondary evaporator **1608** receives inlet air **1628** and outputs first airflow **1634** to primary evaporator **1604**. First airflow **1634**, in general, is at a cooler temperature than inlet air **1628**. To cool incoming inlet air **1628**, secondary evaporator **1608** transfers heat from inlet air **1628** to flow of refrigerant **1626**, thereby causing flow of refrigerant **1626** to evaporate at least partially from liquid to gas.

[0146] Sub-cooling coil **1622**, which is an optional component of dehumidification system **1600**, sub-cools the liquid refrigerant **1626** as it leaves the primary condenser **1606**, the alternate condenser **1620**, or combinations thereof. In embodiments wherein the sub-cooling coil **1622** is disposed within the external condenser unit **1624**, the sub-cooling coil **1622** may receive refrigerant

1626 as it leaves the alternate condenser **1620**, as seen in FIG. **16A**. In embodiments wherein the sub-cooling coil **1622** is disposed adjacent to the primary condenser **1606**, the sub-cooling coil **1622** may receive refrigerant **1626** as it leaves the primary condenser **1606** and/or the alternate condenser **1620**, as seen in FIGS. **16B-16D**. With reference to each of FIGS. **16A-16D**, this, in turn, supplies primary metering device **1614** with a liquid refrigerant that is up to 30 degrees (or more) cooler than before it enters sub-cooling coil **1622**. For example, if flow of refrigerant **1626** entering sub-cooling coil **1622** is 340 psig/105° F./60% vapor, flow of refrigerant **1626** may be 340 psig/80° F./0% vapor as it leaves sub-cooling coil **1622**. The sub-cooled refrigerant **1626** has a greater heat enthalpy factor as well as a greater density, which results in reduced cycle times and frequency of the evaporation cycle of flow of refrigerant **1626**. This results in greater efficiency and less energy use of dehumidification system **1600**.

[0147] Compressor **1612** pressurizes flow of refrigerant **1626**, thereby increasing the temperature of refrigerant **1626**. For example, if flow of refrigerant **1626** entering compressor **1612** is 128 psig/52° F./100% vapor, flow of refrigerant **1626** may be 340 psig/150° F./100% vapor as it leaves compressor **1612**. Compressor **1612** receives flow of refrigerant **1626** from primary evaporator **1604** and supplies the pressurized flow of refrigerant **1626** to modulating valve **1602**.

[0148] Modulating valve **1602** is operable to receive the pressurized flow of refrigerant **1626** from compressor **1612** and to direct the flow of refrigerant to primary condenser **1606**, to alternate condenser **1620**, or to both. In embodiments, the modulating valve **1602** may operate based, at least in part, on a pre-determined temperature set point for the dehumidified airflow **1630** and on an actual temperature of the dehumidified airflow **1630** output by dehumidification system **1600**. Dehumidification system **1600** may utilize modulating valve **1602** to direct heat to be rejected from the flow of refrigerant **1626** away from the primary condenser **1606** and towards the alternate condenser **1620**. Depending on a feedback loop comprising of the pre-determined temperature set point and the actual temperature of the dehumidified airflow **1630**, modulating valve **1602** may be configured to partially open and/or close to direct at least a portion of the flow of refrigerant **1626** to the alternate condenser **1620** and direct a remaining portion of the flow of refrigerant **1626** to the primary condenser **1606**.

[0149] During operation of dehumidification system **1600**, the modulating valve **1602** may direct the flow of refrigerant **1626** to primary condenser **1606** if the temperature of the dehumidified airflow **1630** output by the primary condenser **1606** does not exceed the pre-determined temperature set point monitored by the dehumidification system **1600**. If the temperature of the dehumidified airflow **1630** is greater than the pre-determined temperature set point, the modulating valve **1602** may be actuated to direct at least a portion of the flow of refrigerant **1626** to the alternate condenser **1620** and direct a remaining portion of the flow of refrigerant to the primary condenser **1606**. As the dehumidification system **1600** operates, reduction in the volume of flow of refrigerant **1626** to primary condenser **1606** may reduce the available heat to be rejected into the dehumidified airflow **1630**. With the reduced flow of refrigerant **1626** passing through primary condenser **1606** (for example, the remaining portion of the flow of refrigerant), the rate of heat transfer to the dehumidified airflow **1630** may subsequently be reduced, thereby producing a reduction in the temperature change of an incoming airflow and the output dehumidified airflow **1630**. Once the temperature of the dehumidified airflow **1630** is lower than the pre-determined temperature set point, the modulating valve **1602** may be actuated to direct the at least a portion of the flow of refrigerant **1626** back to the primary condenser **1606**. Any remaining refrigerant **1626** that had been directed to alternate condenser **1620** may combine with the flow of refrigerant **1626** further downstream.

[0150] With reference to FIGS. **16A** and **16B**, alternate condenser **1620** may be disposed in the external condenser unit **1624** and may be any type of coil (e.g., fin tube, micro channel, etc.) operable to receive flow of refrigerant **1626** from modulating valve **1602** and output flow of refrigerant **1626** at a lower temperature. Alternate condenser **1620** transfers heat from flow of

refrigerant **1626**, thereby causing flow of refrigerant **1626** to condense at least partially from gas to liquid. In some embodiments, alternate condenser **1620** completely condenses flow of refrigerant **1626** to a liquid (i.e., 100% liquid). In other embodiments, alternate condenser **1620** partially condenses flow of refrigerant **1626** to a liquid (i.e., less than 100% liquid). As seen in FIG. **16A**, the flow of refrigerant **1626** may be output to sub-cooling coil **1622** disposed adjacent to alternate condenser **1620** within the external condenser unit **1624**. Alternate condenser **1620** and sub-cooling coil **1622** may receive a first outdoor airflow **1640** and output a second outdoor airflow **1642**.

Second outdoor airflow **1642** is, in general, warmer (i.e., have a lower relative humidity) than first outdoor airflow **1640**. In other embodiments, as shown in FIG. **16B**, the first outdoor airflow **1640** may be received by the alternate condenser **1620** without previously flowing through sub-cooling coil **1622**. In FIG. **16B**, the external condenser unit **1624** may include the alternate condenser **1620** and a fan **1644** and may not include the sub-cooling coil **1622**, wherein fan **1644** may be configured to facilitate flow of first outdoor airflow **1640** towards alternate condenser **1620**.

[0151] With reference now to FIGS. **16C-16D**, alternate condenser **1620** may be any type of liquid-cooled heat exchanger operable to transfer heat from the flow of refrigerant **1626** to the flow of a fluid **1646**, wherein the alternate condenser **1620** receives flow of refrigerant **1626** from modulating valve **1602** and outputs flow of refrigerant **1626** to sub-cooling coil **1622**. In embodiments, the fluid **1646** may be any suitable fluid, such as water or a mixture of water and glycol. Alternate condenser **1620** receives both the flow of fluid **1646** and the flow of refrigerant **1626** during operation of dehumidification system **1600**, wherein the alternate condenser **1620** is operable to transfer heat from the flow of refrigerant **1626**, thereby causing flow of refrigerant **1626** to condense at least partially from gas to liquid. In some embodiments, alternate condenser **1620** completely condenses flow of refrigerant **1626** to a liquid (i.e., 100% liquid). In other embodiments, alternate condenser **1620** partially condenses flow of refrigerant **1626** to a liquid (i.e., less than 100% liquid).

[0152] As illustrated in FIGS. **16C-16D**, the dehumidification system **1600** may further comprise a first water pump **1648**. The first water pump **1648** may be disposed external to the alternate condenser **1620**. The first water pump may be any suitable device operable to provide for the flow of fluid **1646**. As depicted in FIG. **16C**, the first water pump **1648** may be disposed at any suitable location between the alternate condenser **1620** and a heat exchanger **1654** operable to cycle the flow of fluid **1646** between the heat exchanger **1654** and the alternate condenser **1620**. As depicted in FIG. **16D**, the first water pump **1648** may be disposed at any suitable location between the alternate condenser **1620** and an external source **1652** operable to cycle the flow of fluid **1646** between the external source **1652** and the alternate condenser **1620**.

[0153] With reference to FIG. **16C**, heat exchanger **1654** may receive the flow of fluid **1646** from alternate condenser **1620** and output flow of fluid **1646** after transferring heat away from the flow of fluid **1646**. Heat exchanger **1654** may be any suitable type of heat exchanger, such as a cooling tower or a dry cooler. Heat exchanger **1654** receives the flow of fluid **1646** and a first outdoor airflow **1656**, wherein heat is transferred between the flow of fluid **1646** and the first outdoor airflow **1656**. Heat exchanger **1654** may further output the flow of fluid **1646** and a second outdoor airflow **1658**, wherein the flow of fluid **1646** leaving the heat exchanger **1654** is at a lower temperature than the flow of fluid **1646** received by the heat exchanger **1654**, and the second outdoor airflow **1658** is at a greater temperature than the first outdoor airflow **1654**.

[0154] In embodiments wherein the heat exchanger **1654** is a cooling tower, the heat exchanger **1654** may be operable to dispense the flow of fluid **1646** within its internal structure, wherein the fluid **1646** directly contacts the first outdoor airflow **1656** as the fluid **1646** flows through the heat exchanger **1654** and transfers heat to the first outdoor airflow **1656**. At least a portion of the fluid **1646** may evaporate and exit to the atmosphere as the heat transfers from the fluid **1646** to the first outdoor airflow **1656**, and the heat exchanger **1654** may collect a remaining portion of the fluid **1646** after transferring heat to the first outdoor airflow **1656**, wherein the remaining portion of the

fluid **1646** is at a lower temperature. In embodiments wherein the heat exchanger **1654** is a dry cooler, the heat exchanger **1654** may be operable to induce the first outdoor airflow **1656** to flow through the heat exchanger **1654** where heat transfers indirectly between the first outdoor airflow **1656** and the flow of fluid **1646**. In these embodiments, heat transfer would not result in loss of a portion of the fluid **1646** through evaporation to the atmosphere.

[0155] With reference to FIG. **16D**, external source **1652** may receive the flow of fluid **1646** and output flow of fluid **1646** to the alternate condenser **1620** via first water pump **1648**. External source **1652** may be configured to contain and/or store a volume of fluid **1646** to be used by alternate condenser **1620** to lower the temperature of the flow of refrigerant **1626** in the dehumidification system **1600**. Without limitations, the external source **1652** may be selected from a group consisting of a ground reservoir, a natatorium, an outdoor body of water, and any combinations thereof. In embodiments wherein the external source **1652** is a ground reservoir, the external source **1652** may implement an open or closed ground water system, wherein the conduit providing for the flow of fluid **1646** within the ground reservoir may be disposed substantially parallel to a horizontal plane of the ground surface, substantially perpendicular to the horizontal plane of the ground surface, or combinations thereof.

[0156] In embodiments wherein the external source **1652** is a natatorium, the external source **1652** may be within a multi-loop system operable to contain and cool the flow of fluid **1646** before the alternate condenser **1620** uses the flow of fluid **1646** to lower the temperature of the flow of refrigerant **1626**. The external source **1652** may be configured to receive the flow of fluid **1646** from alternate condenser **1620** at a first temperature and output flow of fluid **1646** to alternate condenser **1620** at a second temperature after transferring heat away from the flow of fluid **1646**, wherein the second temperature is lower than the first temperature. External source **1652** receives the flow of fluid **1646** and may receive a flow of a secondary fluid (not shown), wherein heat is transferred between the flow of fluid **1646** and the flow of secondary fluid. External source **1652** may then output the flow of fluid **1646** and the flow of secondary fluid, wherein the flow of fluid **1646** leaving the external source **1652** is at a lower temperature than the flow of fluid **1646** received by the external source **1652**, and wherein the flow of secondary fluid leaving the external source **1652** is at a greater temperature than the flow of secondary fluid received by the external source **1652**.

[0157] The flow of secondary fluid may then be directed to a tertiary condenser (not shown). The tertiary condenser receives the flow of secondary fluid from external source **1652** and outputs flow of secondary fluid back to the external source **1652** at a lower temperature. The tertiary condenser may be any type of air-cooled or liquid-cooled heat exchanger operable to transfer heat away from the flow of secondary fluid. In embodiments, a second pump (not shown) may be at any suitable position in relation to the external source **1652** and the tertiary condenser operable to cycle the flow of secondary fluid between the external source **1652** and the tertiary condenser, wherein the second pump may be any suitable device operable to provide for the flow of secondary fluid.

[0158] Referring back to each of FIGS. **16A-16D**, fan **1618** may include any suitable components operable to draw inlet air **1628** into dehumidification system **1600** and through secondary evaporator **1608**, primary evaporator **1604**, secondary condenser **1610**, sub-cooling coil **1622**, and primary condenser **1606**. Fan **1618** may be any type of air mover (e.g., axial fan, forward inclined impeller, and backward inclined impeller, etc.). For example, fan **1618** may be a backward inclined impeller positioned adjacent to primary condenser **1606** as illustrated in FIGS. **16A-16D**. While fan **1618** is depicted in FIGS. **16A-16D** as being located adjacent to primary condenser **1606**, it should be understood that fan **1618** may be located anywhere along the airflow path of dehumidification system **1600**. For example, fan **1618** may be positioned in the airflow path of any one of airflows **1628**, **1634**, **1632**, **1636**, **1638**, or **1630**. Moreover, dehumidification system **1600** may include one or more additional fans positioned within any one or more of these airflow paths. Similarly, with reference to FIGS. **16A-16B**, while a fan **1644** of external condenser unit **1624** is depicted as being

located above alternate condenser **1620**, it should be understood that fan **1644** may be located anywhere (e.g., above, below, beside) with respect to alternate condenser **1620** and optional sub-cooling coil **1622**, so long as fan **1644** is appropriately positioned and configured to facilitate flow of first outdoor airflow **1640** towards alternate condenser **1620**.

[0159] Primary metering device **1614** and secondary metering device **1616** are any appropriate type of metering/expansion device. In some embodiments, primary metering device **1614** is a thermostatic expansion valve (TXV) and secondary metering device **1616** is a fixed orifice device (or vice versa). In certain embodiments, metering devices **1614** and **1616** remove pressure from flow of refrigerant **1626** to allow expansion or change of state from a liquid to a vapor in evaporators **1604** and **1608**. The high-pressure liquid (or mostly liquid) refrigerant entering metering devices **1614** and **1616** is at a higher temperature than the liquid refrigerant **1626** leaving metering devices **1614** and **1616**. For example, if flow of refrigerant **1626** entering primary metering device **1614** is 340 psig/80° F./0% vapor, flow of refrigerant **1626** may be 196 psig/68° F./5% vapor as it leaves primary metering device **1614**. As another example, if flow of refrigerant **1626** entering secondary metering device **1616** is 196 psig/68° F./4% vapor, flow of refrigerant **1626** may be 128 psig/44° F./14% vapor as it leaves secondary metering device **1616**.

[0160] Refrigerant **1626** may be any suitable refrigerant such as R410a. In general, dehumidification system **1600** utilizes a closed refrigeration loop of refrigerant **1626** that passes from compressor **1612** through modulating valve **1602**, primary condenser **1612** and/or alternate condenser **1620**, (optionally) sub-cooling coil **1622**, primary metering device **1614**, secondary evaporator **1608**, secondary condenser **1610**, secondary metering device **1616**, and primary evaporator **1604**. Compressor **1612** pressurizes flow of refrigerant **1626**, thereby increasing the temperature of refrigerant **1626**. Primary and secondary condensers **1606** and **1610**, which may include any suitable heat exchangers, cool the pressurized flow of refrigerant **1626** by facilitating heat transfer from the flow of refrigerant **1626** to the respective airflows passing through them (i.e., third or fourth airflow **1636**, **1638** and second airflow **1632**). Further, alternate condenser **1620**, which may include any suitable heat exchanger, cools the pressurized flow of refrigerant **1626** by facilitating heat transfer from the flow of refrigerant **1626** to either the airflow passing through it (i.e., first outdoor airflow **1640** as illustrated in FIGS. **16A-16B**) or to the flow of fluid provided by the external source **1652** passing through it (i.e., flow of fluid **1646** as illustrated in FIGS. **16C-16D**). The cooled flow of refrigerant **1626** leaving primary and/or alternate condensers **1606** and **1620** may enter primary metering device **1614**, which is operable to reduce the pressure of flow of refrigerant **1626**, thereby reducing the temperature of flow of refrigerant **1626**. The cooled flow of refrigerant **1626** leaving secondary condenser **1610** may enter secondary metering device **1616**, which is operable to reduce the pressure of flow of refrigerant **1626**, thereby reducing the temperature of flow of refrigerant **1626**. Primary and secondary evaporators **1604** and **1608**, which may include any suitable heat exchanger, receive flow of refrigerant **1626** from secondary metering device **1616** and primary metering device **1614**, respectively. Primary and secondary evaporators **1604** and **1608** facilitate the transfer of heat from the respective airflows passing through them (i.e., inlet air **1628** and first airflow **1634**) to flow of refrigerant **1626**. Flow of refrigerant **1626**, after leaving primary evaporator **1604**, passes back to compressor **1612**, and the cycle is repeated.

[0161] In certain embodiments, the above-described refrigeration loop may be configured such that evaporators **1604** and **1608** operate in a flooded state. In other words, flow of refrigerant **1626** may enter evaporators **1604** and **1608** in a liquid state, and a portion of flow of refrigerant **1626** may still be in a liquid state as it exits evaporators **1604** and **1608**. Accordingly, the phase change of flow of refrigerant **1626** (liquid to vapor as heat is transferred to flow of refrigerant **1626**) occurs across evaporators **1604** and **1608**, resulting in nearly constant pressure and temperature across the entire evaporators **1604** and **1608** (and, as a result, increased cooling capacity).

[0162] In operation of example embodiments of dehumidification system **1600**, inlet air **1628** may be drawn into dehumidification system **1600** by fan **1618**. Inlet air **1628** passes through secondary

evaporator **1608** in which heat is transferred from inlet air **1628** to the cool flow of refrigerant **1626** passing through secondary evaporator **1608**. As a result, inlet air **1628** may be cooled. As an example, if inlet air **1628** is 80° F./60% humidity, secondary evaporator **1608** may output first airflow **1634** at 70° F./84% humidity. This may cause flow of refrigerant **1626** to partially vaporize within secondary evaporator **1608**. For example, if flow of refrigerant **1626** entering secondary evaporator **1608** is 196 psig/68° F./5% vapor, flow of refrigerant **1626** may be 196 psig/68° F./38% vapor as it leaves secondary evaporator **1608**.

[0163] The cooled inlet air **1628** leaves secondary evaporator **1608** as first airflow **1634** and enters primary evaporator **1604**. Like secondary evaporator **1608**, primary evaporator **1604** transfers heat from first airflow **1634** to the cool flow of refrigerant **1626** passing through primary evaporator **1604**. As a result, first airflow **1634** may be cooled to or below its dew point temperature, causing moisture in first airflow **1634** to condense (thereby reducing the absolute humidity of first airflow **1634**). As an example, if first airflow **1634** is 70° F./84% humidity, primary evaporator **1604** may output second airflow **1632** at 54° F./98% humidity. This may cause flow of refrigerant **1626** to partially or completely vaporize within primary evaporator **1604**. For example, if flow of refrigerant **1626** entering primary evaporator **1604** is 128 psig/44° F./14% vapor, flow of refrigerant **1626** may be 128 psig/52° F./100% vapor as it leaves primary evaporator **1604**.

[0164] The cooled first airflow **1634** leaves primary evaporator **1604** as second airflow **1632** and enters secondary condenser **1610**. Secondary condenser **1610** facilitates heat transfer from the hot flow of refrigerant **1626** passing through the secondary condenser **1610** to second airflow **1632**. This reheats second airflow **1632**, thereby decreasing the relative humidity of second airflow **1632**. As an example, if second airflow **1632** is 54° F./98% humidity, secondary condenser **1610** may output third airflow **1636** at 65° F./68% humidity. This may cause flow of refrigerant **1626** to partially or completely condense within secondary condenser **1610**. For example, if flow of refrigerant **1626** entering secondary condenser **1610** is 196 psig/68° F./38% vapor, flow of refrigerant **1626** may be 196 psig/68° F./4% vapor as it leaves secondary condenser **1610**.

[0165] In some embodiments, the dehumidified second airflow **1632** leaves secondary condenser **1610** as third airflow **1636** and enters primary condenser **1606**, as illustrated in FIG. 16A. Primary condenser **1606** facilitates heat transfer from the hot flow of refrigerant **1626** passing through the primary condenser **1606** to third airflow **1636**. This further heats third airflow **1636**, thereby further decreasing the relative humidity of third airflow **1636**. As an example, if third airflow **1636** is 65° F./68% humidity, primary condenser **1606** may output dehumidified air **1630** at 102° F./19% humidity. This may cause flow of refrigerant **1626** to partially or completely condense within primary condenser **1606**. For example, if flow of refrigerant **1626** entering primary condenser **1606** is 340 psig/150° F./100% vapor, flow of refrigerant **1626** may be 340 psig/105° F./60% vapor as it leaves primary condenser **1606**.

[0166] As described above, some embodiments of dehumidification system **1600** may include a sub-cooling coil **1622** in the airflow between secondary condenser **1610** and primary condenser **1606**, as best seen in FIGS. 16B-16D. Sub-cooling coil **1622** facilitates heat transfer from the hot flow of refrigerant **1626** passing through sub-cooling coil **1622** to third airflow **1636**. This further heats third airflow **1636**, thereby further decreasing the relative humidity of third airflow **1636**. As an example, if third airflow **1636** is 65° F./68% humidity, sub-cooling coil **1622** may output fourth airflow **1638** at 81° F./37% humidity. This may cause flow of refrigerant **1626** to partially or completely condense within sub-cooling coil **1622**. For example, if flow of refrigerant **1626** entering sub-cooling coil **1622** is 340 psig/150° F./60% vapor, flow of refrigerant **1626** may be 340 psig/80° F./0% vapor as it leaves sub-cooling coil **1622**. In these embodiments, the fourth airflow **1638** may then undergo heat transfer in primary condenser **1606** to produce dehumidified airflow **1630**.

[0167] Some embodiments of dehumidification system **1600** may include a controller that may include one or more computer systems at one or more locations. Each computer system may

include any appropriate input devices (such as a keypad, touch screen, mouse, or other device that can accept information), output devices, mass storage media, or other suitable components for receiving, processing, storing, and communicating data. Both the input devices and output devices may include fixed or removable storage media such as a magnetic computer disk, CD-ROM, or other suitable media to both receive input from and provide output to a user. Each computer system may include a personal computer, workstation, network computer, kiosk, wireless data port, personal data assistant (PDA), one or more processors within these or other devices, or any other suitable processing device. In short, the controller may include any suitable combination of software, firmware, and hardware.

[0168] The controller may additionally include one or more processing modules. Each processing module may each include one or more microprocessors, controllers, or any other suitable computing devices or resources and may work, either alone or with other components of dehumidification system **1600**, to provide a portion or all of the functionality described herein. The controller may additionally include (or be communicatively coupled to via wireless or wireline communication) computer memory. The memory may include any memory or database module and may take the form of volatile or non-volatile memory, including, without limitation, magnetic media, optical media, random access memory (RAM), read-only memory (ROM), removable media, or any other suitable local or remote memory component.

[0169] Although particular implementations of dehumidification system **1600** are illustrated and primarily described, the present disclosure contemplates any suitable implementation of dehumidification system **1600**, according to particular needs. Moreover, although various components of dehumidification system **1600** have been depicted as being located at particular positions and relative to one another, the present disclosure contemplates those components being positioned at any suitable location, according to particular needs.

[0170] FIGS. **17A**, **17B**, and **17C** illustrate an example dehumidification system **1700** with a first modulating valve **1702** that may be used to control the ratio of sensible heat to latent heat for the surrounding airflow. First modulating valve **1702** may be configured to actuate between different modes of operation based on input received from a superheat control evaporator **1704**. For example, operations of dehumidification system **1700** may be based, at least in part, on controlling the superheat from one or more evaporator coils of the dehumidification system **1700**. Providing increased superheat control may prevent the cooling of an airflow already determined to be at a control or set temperature.

[0171] Dehumidification system **1700** may comprise the first modulating valve **1702**, the superheat control evaporator **1704**, a primary evaporator **1706**, a primary condenser **1708**, a secondary evaporator **1710**, a secondary condenser **1712**, a compressor **1714**, a primary metering device **1716**, a secondary metering device **1718**, a fan **1720**, a second modulating valve **1722**, and an alternate condenser **1724**. As illustrated, the alternate condenser **1724** may be disposed in an external condenser unit **1726** and may be air-cooled. In alternate embodiments, the alternate condenser **1724** may be liquid-cooled. In some embodiments, dehumidification system **1700** may additionally include an optional sub-cooling coil **1728**. With reference to the figures, the optional sub-cooling coil **1728** may be disposed adjacent to the primary condenser **1708**, wherein sub-cooling coil **1728** and primary condenser **1708** may be combined into a single coil. FIGS. **17A** and **17B** illustrate an embodiment of dehumidification system **1700** wherein superheat control evaporator **1704** is disposed separately from secondary evaporator **1710**, and FIG. **17C** illustrates an embodiment of dehumidification system **1700** wherein superheat control evaporator **1704** and secondary evaporator **1710** are jointly integrated into an intermixed coil unit **1730**.

[0172] With reference to each of FIGS. **17A-17C**, a flow of refrigerant **1732** is circulated through dehumidification system **1700**, as illustrated. In general, dehumidification system **1700** may receive one or more inlet airflows **1734**, remove water from the one or more inlet airflows **1734**, and output a dischargeable airflow **1736**. In embodiments, dischargeable airflow **1736** may be at

least partially dehumidified and/or at a lower temperature than the one or more inlet airflows **1734**. Water may be removed from the one or more inlet airflows **1734** using a refrigeration cycle of flow of refrigerant **1732**. By including secondary evaporator **1710** and secondary condenser **1712**, however, dehumidification system **1700** may cause at least part of the flow of refrigerant **1732** to evaporate and condense twice in a single refrigeration cycle. This may increase the refrigeration capacity over typical systems without adding any additional power to the compressor, thereby increasing the overall dehumidification efficiency of the system.

[0173] In general, dehumidification system **1700** may attempt to match the saturating temperature of secondary evaporator **1710** to the saturating temperature of secondary condenser **1712**. The saturating temperature of secondary evaporator **1710** and secondary condenser **1712** generally is controlled according to the equation: (temperature of inlet airflow **1734**+temperature of an airflow received by secondary condenser **1712**)/2. As the saturating temperature of secondary evaporator **1710** is lower than inlet airflow **1734**, evaporation happens in secondary evaporator **1710**. As the saturating temperature of secondary condenser **1712** is higher than an airflow received by secondary condenser **1712**, condensation happens in the secondary condenser **1712**. The amount of refrigerant **1732** evaporating in secondary evaporator **1710** may be substantially equal to that condensing in secondary condenser **1712**.

[0174] Superheat control evaporator **1704** may receive a flow of refrigerant **1732** from primary evaporator **1706** and may output a flow of refrigerant **1732** to compressor **1714**. Superheat control evaporator **1704** may be any type of coil (e.g., fin tube, micro channel, etc.). With reference to FIG. **17A**, superheat control evaporator **1704** may receive inlet airflow **1734** and may output a first airflow **1738** to secondary evaporator **1710**. First airflow **1738**, in general, is at a cooler temperature than inlet airflow **1734**. To cool incoming inlet airflow **1734**, superheat control evaporator **1734** may transfer heat from inlet airflow **1734** to the flow of refrigerant **1732**, thereby causing the flow of refrigerant **1732** to evaporate at least partially from liquid to gas.

[0175] Secondary evaporator **1710** may receive the flow of refrigerant **1732** from primary metering device **1716** and may output a flow of refrigerant **1732** to the first modulating valve **1702**. Similar to superheat control evaporator **1704**, secondary evaporator **1710** may be any type of coil (e.g., fin tube, micro channel, etc.). Secondary evaporator **1710** receives first airflow **1738** and outputs a second airflow **1740** to primary evaporator **1706**. Second airflow **1740**, in general, is at a cooler temperature than first airflow **1738**. To cool incoming first airflow **1738**, secondary evaporator **1710** transfers heat from first airflow **1738** to flow of refrigerant **1732**, thereby causing flow of refrigerant **1732** to evaporate at least partially from liquid to gas.

[0176] In this embodiment, superheat control evaporator **1704** may be disposed upstream of both the secondary evaporator **1710** and primary evaporator **1706** with reference to the air flow through the dehumidification system **1700**. For example, the superheat control evaporator **1704** may be disposed in series with the other coils of dehumidification system **1700** and may be the first component to receive one or more inlet airflows **1734**. In other embodiments, superheat control evaporator **1704** may be disposed in various other configurations in view of the other components of dehumidification system **1700**.

[0177] For example, with reference now to FIG. **17B**, superheat control evaporator **1704** may be disposed parallel to the secondary evaporator **1710**. Superheat control evaporator **1704** and secondary evaporator **1710** may receive one or more inlet airflows **1734** and may discharge treated air to primary evaporator **1706**. In this embodiment, superheat control evaporator **1704** may receive a first inlet airflow **1734a**, and secondary evaporator **1710** may receive a second inlet airflow **1734b**. Superheat control evaporator **1704** may transfer heat from first inlet airflow **1734a** to the flow of refrigerant **1732** flowing through the superheat control evaporator **1704** and output the generated first airflow **1738** to primary evaporator **1706**. Similarly, secondary evaporator **1710** may transfer heat from second inlet airflow **1734b** to the flow of refrigerant **1732** flowing through the secondary evaporator **1710** and output second airflow **1740** to primary evaporator **1706**. As

illustrated, primary evaporator **1706** may receive both the first airflow **1738** and the second airflow **1740** for further operations within dehumidification system **1700**. The present embodiment may provide the superheat control evaporator **1704** and secondary evaporator **1710** being disposed separate from each other and receiving separate inlet airflows **1734**. Further, there may be separate ductwork and/or conduits directing the airflows received by the dehumidification system **1700** (i.e., inlet airflows **1734a,b**) towards the primary evaporator **1706**.

[0178] FIG. **17C** illustrates another embodiment of dehumidification system **1700**, wherein both the superheat control evaporator **1704** and secondary evaporator **1710** are jointly integrated into the intermixed coil unit **1730**. In certain embodiments, the superheat control evaporator **1704** and secondary evaporator **1710** may be collectively referred to as an “intermixed coil unit **1730**” when coupled together. The intermixed coil unit **1730** may comprise any suitable size, height, shape, and any combinations thereof. The intermixed coil unit **1730** may further comprise any suitable housing or containment equipment for the superheat control evaporator **1704** and the secondary evaporator **1710**. For example, the superheat control evaporator **1704** may be physically coupled or secured to the secondary evaporator **1710** in order for both evaporator coils **1704**, **1710** to receive the same inlet airflow **1734**. While both the superheat control evaporator **1704** and secondary evaporator **1710** may be coupled together, there may be separate flowpaths for the refrigerant **1732** flowing through superheat control evaporator **1704** and secondary evaporator **1710** (as illustrated). In embodiments, the intermixed coil unit **1730** may receive the inlet airflow **1734** and may output the first airflow **1738** to the primary evaporator **1706**. The first airflow **1738** may be generated by transferring heat from the inlet airflow **1734** to the flow of refrigerant **1732** within both the superheat control evaporator **1704** and secondary evaporator **1710** as the inlet airflow **1734** passes through both the superheat control evaporator **1704** and the secondary evaporator **1710**.

[0179] Referring back to each of FIGS. **17A-17C**, the primary evaporator **1706** may receive a flow of refrigerant **1732** and may output the flow of refrigerant **1732** to the superheat control evaporator **1704**. As illustrated, the primary evaporator **1706** may receive the flow of refrigerant **1732** from secondary metering device **1718**, wherein the secondary metering device **1718** may receive the flow of refrigerant **1732** from the secondary condenser **1712** and/or from the first modulating valve **1702**. In other embodiments, the primary evaporator **1706** may receive the flow of refrigerant **1732** from the first modulating valve **1702**, wherein the first modulating valve **1702** may direct refrigerant to bypass both secondary condenser **1712** and secondary metering device **1718**, wherein the output of the first modulating valve **1702** is connected to a location downstream of the secondary metering device **1718** but upstream of the primary evaporator **1706**.

[0180] Primary evaporator **1706** may be any type of coil (e.g., fin tube, micro channel, etc.). Primary evaporator **1706** may be configured to receive first airflow **1738** and/or second airflow **1740** and generate a third airflow **1742** to be discharged. For example, with reference to FIG. **17A**, primary evaporator **1706** may receive second airflow **1740** from secondary evaporator **1710**. Referring to FIG. **17B**, primary evaporator **1706** may receive second airflow **1740** from secondary evaporator **1710** and may receive first airflow **1738** from superheat control evaporator **1704**. With reference to FIG. **17C**, primary evaporator **1706** may receive first airflow **1738** from intermixed coil unit **1730**. Third airflow **1742**, in general, is at a cooler temperature than first airflow **1738** and/or second airflow **1740**. In embodiments, primary evaporator **1706** may transfer heat from first airflow **1738** and/or second airflow **1740** to the flow of refrigerant **1732**, thereby causing flow of refrigerant **1732** to evaporate at least partially from liquid to gas. This transfer of heat from first airflow **1738** and/or second airflow **1740** to flow of refrigerant **1732** may further remove water from first airflow **1738** and/or second airflow **1740**.

[0181] Secondary condenser **1712** may receive the flow of refrigerant **1732** from the first modulating valve **1702** and may output the flow of refrigerant **1732** to secondary metering device **1718**. Secondary condenser **1712** may be any type of coil (e.g., fin tube, micro channel, etc.). Secondary condenser **1712** may receive the third airflow **1742** from primary evaporator **1706** and

may output a fourth airflow **1744**. Fourth airflow **1744** may be, in general, warmer and drier (i.e., the dew point may be the same but relative humidity may be lower) than third airflow **1742**. Secondary condenser **1712** may generate fourth airflow **1744** by transferring heat from flow of refrigerant **1732** to third airflow **1742**, thereby causing flow of refrigerant **1732** to condense at least partially from gas to liquid.

[0182] First modulating valve **1702** may be configured to receive the flow of refrigerant **1732** from secondary evaporator **1710** and to direct the flow of refrigerant **1732** to secondary condenser **1712**, to primary evaporator **1706**, or to both. In embodiments, the first modulating valve **1702** may operate based, at least in part, on the superheat measured at one or more of the evaporator coils within dehumidification system **1700**, such as at superheat control evaporator **1704**.

Dehumidification system **1700** may utilize first modulating valve **1702** to direct the flow of refrigerant **1732** to secondary condenser **1712**, to bypass the secondary condenser **1712** and towards the primary evaporator **1706**, or a combination thereof. Depending on a feedback loop, first modulating valve **1702** may be configured to partially open and/or close to direct at least a portion of the flow of refrigerant **1732** to the secondary condenser **1712** and direct a remaining portion of the flow of refrigerant **1732** to the primary evaporator **1706**.

[0183] In embodiments, dehumidification system **1700** may operate in a first mode of operation. During the first mode of operation, the first modulating valve **1702** may be actuated to direct the flow of refrigerant **1732** to secondary condenser **1712**. As the refrigerant **1732** flows through the secondary condenser **1712**, the secondary condenser **1712** may generate the fourth airflow **1744**. The dehumidification system **1700** may operate in the first mode of operation to dehumidify or remove water from the air to be output as the dischargeable airflow **1736**. In further embodiments, dehumidification system **1700** may operate in a second mode of operation. During the second mode of operation, the first modulating valve **1702** may be actuated to direct the flow of refrigerant **1732** to primary evaporator **1706**, thereby bypassing the secondary condenser **1712**. As the refrigerant **1732** does not flow through the secondary condenser **1712**, the secondary condenser **1712** may not be capable of transferring heat between the refrigerant **1732** and the received third airflow **1742**. As a result, the resulting airflow passing through the secondary condenser **1712** (i.e., the fourth airflow **1744**) may comprise approximately the same temperature and humidity as the third airflow **1742**. The dehumidification system **1700** may operate in the second mode of operation to lower the temperature of the air to be output as the dischargeable airflow **1736** and not to dehumidify. In other embodiments, the dehumidification system **1700** may operate in a third mode of operation, wherein the first modulating valve **1702** is operable to direct a portion of the flow of refrigerant **1732** to the secondary condenser **1712** and a remaining portion of the flow of refrigerant **1732** to the primary evaporator **1706**. As at least a portion of the refrigerant **1732** flows through the secondary condenser **1712**, the secondary condenser **1712** may generate the fourth airflow **1744** by transferring heat from that portion of refrigerant **1732** to the received third airflow **1742**. In embodiments, the fourth airflow **1744** of the third mode of operation may be more humid than the fourth airflow **1744** of the first mode of operation and less humid than the fourth airflow **1744** of the second mode of operation. Further, the fourth airflow **1744** of the third mode of operation may be at a greater temperature than the fourth airflow **1744** of the second mode of operation and at a lower temperature than the fourth airflow **1744** of the first mode of operation.

[0184] The primary condenser **1708** may be any type of coil (e.g., fin tube, micro channel, etc.). Primary condenser **1708** is operable to receive the flow of refrigerant **1732** from the second modulating valve **1722** and output the flow of refrigerant **1732** to either primary metering device **1716** or sub-cooling coil **1728**. As illustrated, primary condenser **1708** may output the flow of refrigerant **1732** to the optional sub-cooling coil **1728** before the flow of refrigerant **1732** flows to primary metering device **1716**. In these embodiments, sub-cooling coil **1728** may be optional for dehumidification system **1700**, and primary condenser **1708** may alternatively direct the flow of refrigerant **1732** to the primary metering device **1716**. Primary condenser **1708** may be configured

to receive a fifth airflow **1746** generated by the sub-cooling coil **1728** and output dischargeable airflow **1736**. With reference to each of FIGS. **17A-17C**, dischargeable airflow **1736** may be, in general, warmer and drier (i.e., have a lower relative humidity) than either fourth airflow **1744** or fifth airflow **1746**. Primary condenser **1708** may generate dischargeable airflow **1736** by transferring heat away from flow of refrigerant **1732**, thereby causing flow of refrigerant **1732** to condense at least partially from gas to liquid. In some embodiments, primary condenser **1708** completely condenses flow of refrigerant **1732** to a liquid (i.e., 100% liquid). In other embodiments, primary condenser **1708** partially condenses flow of refrigerant **1732** to a liquid (i.e., less than 100% liquid).

[0185] Sub-cooling coil **1728**, which is an optional component of dehumidification system **1700**, may be configured to sub-cool the liquid refrigerant **1732** as it leaves the primary condenser **1708**, the alternate condenser **1724**, or combinations thereof. In embodiments wherein the sub-cooling coil **1728** is disposed adjacent to the primary condenser **1708**, the sub-cooling coil **1728** may receive refrigerant **1732** as it leaves the primary condenser **1708** and/or the alternate condenser **1724**, as seen in FIGS. **17A-17C**. This, in turn, may supply primary metering device **1716** with a liquid refrigerant that is up to 30 degrees (or more) cooler than before it enters sub-cooling coil **1728**. For example, if flow of refrigerant **1732** entering sub-cooling coil **1728** is 340 psig/105° F./60% vapor, flow of refrigerant **1732** may be 340 psig/80° F./0% vapor as it leaves sub-cooling coil **1728**. The sub-cooled refrigerant **1732** has a greater heat enthalpy factor as well as a greater density, which results in reduced cycle times and frequency of the evaporation cycle of flow of refrigerant **1732**. This may result in greater efficiency and less energy use of dehumidification system **1700**.

[0186] Compressor **1714** may be configured to pressurize the flow of refrigerant **1732**, thereby increasing the temperature of refrigerant **1732**. For example, if flow of refrigerant **1732** entering compressor **1714** is 128 psig/52° F./100% vapor, flow of refrigerant **1732** may be 340 psig/150° F./100% vapor as it leaves compressor **1714**. Compressor **1714** may be configured to receive flow of refrigerant **1732** from superheat control evaporator **1704** and to supply the pressurized flow of refrigerant **1732** to the second modulating valve **1722**.

[0187] Second modulating valve **1722** may be operable to receive the pressurized flow of refrigerant **1732** from compressor **1714** and to direct the flow of refrigerant **1732** to primary condenser **1708**, to alternate condenser **1724**, or to both. In embodiments, the second modulating valve **1722** may operate based, at least in part, on a pre-determined temperature set point for the dischargeable airflow **1736** and on an actual temperature of the dischargeable airflow **1736** output by dehumidification system **1700**. Dehumidification system **1700** may utilize second modulating valve **1722** to direct heat to be rejected from the flow of refrigerant **1732** away from the primary condenser **1708** and towards the alternate condenser **1724**. Depending on a feedback loop comprising of the pre-determined temperature set point and the actual temperature of the dischargeable airflow **1736**, second modulating valve **1722** may be configured to partially open and/or close to direct at least a portion of the flow of refrigerant **1732** to the alternate condenser **1724** and direct a remaining portion of the flow of refrigerant **1732** to the primary condenser **1708**.

[0188] During operation of dehumidification system **1700**, the second modulating valve **1722** may direct the flow of refrigerant **1732** to primary condenser **1708** if the temperature of the dischargeable airflow **1736** output by the primary condenser **1708** does not exceed the pre-determined temperature set point monitored by the dehumidification system **1700**. If the temperature of the dischargeable airflow **1736** is greater than the pre-determined temperature set point, the second modulating valve **1722** may be actuated to direct at least a portion of the flow of refrigerant **1732** to the alternate condenser **1724** and direct a remaining portion of the flow of refrigerant **1732** to the primary condenser **1708**. As the dehumidification system **1700** operates, reduction in the volume of flow of refrigerant **1732** to primary condenser **1708** may reduce the available heat to be rejected into the dischargeable airflow **1736**. With the reduced flow of

refrigerant **1732** passing through primary condenser **1708** (for example, the remaining portion of the flow of refrigerant), the rate of heat transfer to the dischargeable airflow **1736** may subsequently be reduced, thereby producing a reduction in the temperature change of an incoming airflow and the output dischargeable airflow **1736**. Once the temperature of the dischargeable airflow **1736** is lower than the pre-determined temperature set point, the second modulating valve **1722** may be actuated to direct the at least a portion of the flow of refrigerant **1732** back to the primary condenser **1708**. Any remaining refrigerant **1732** that had been directed to alternate condenser **1724** may combine with the flow of refrigerant **1732** further downstream.

[0189] As illustrated, alternate condenser **1724** may be disposed in the external condenser unit **1726** and may be any type of coil (e.g., fin tube, micro channel, etc.) operable to receive flow of refrigerant **1732** from second modulating valve **1722** and output flow of refrigerant **1732** at a lower temperature. Alternate condenser **1724** may be configured to transfer heat away from flow of refrigerant **1732**, thereby causing flow of refrigerant **1732** to condense at least partially from gas to liquid. In some embodiments, alternate condenser **1724** completely condenses flow of refrigerant **1732** to a liquid (i.e., 100% liquid). In other embodiments, alternate condenser **1724** partially condenses flow of refrigerant **1732** to a liquid (i.e., less than 100% liquid). Alternate condenser **1724** may receive a first outdoor airflow **1748** and output a second outdoor airflow **1750**. Second outdoor airflow **1750** is, in general, warmer (i.e., have a lower relative humidity) than first outdoor airflow **1748**. As illustrated, the external condenser unit **1726** may include the alternate condenser **1724** and a fan **1752**, wherein fan **1752** may be configured to facilitate flow of first outdoor airflow **1748** towards alternate condenser **1724**. While alternate condenser **1724** may be air-cooled, alternate condenser **1724** may alternatively be liquid-cooled. In one or more embodiments, alternate condenser **1724** may be any type of liquid-cooled heat exchanger operable to transfer heat from the flow of refrigerant **1732** to the flow of a suitable fluid, such as water or a mixture of water and glycol.

[0190] Referring back to each of FIGS. **17A-17C**, fan **1720** may include any suitable components operable to draw one or more inlet airflows **1734** into dehumidification system **1700** and through superheat control evaporator **1704**, secondary evaporator **1710**, primary evaporator **1706**, secondary condenser **1712**, sub-cooling coil **1728**, and primary condenser **1708**. Fan **1720** may be any type of air mover (e.g., axial fan, forward inclined impeller, and backward inclined impeller, etc.). For example, fan **1720** may be a backward inclined impeller positioned adjacent to primary condenser **1720** as illustrated in FIGS. **17A-17C**. While fan **1720** is depicted as being located adjacent to primary condenser **1708**, it should be understood that fan **1720** may be located anywhere along the airflow path of dehumidification system **1700**. For example, fan **1720** may be positioned in the airflow path of any one of airflows **1734**, **1738**, **1740**, **1742**, **1744**, **1746**, or **1736**. Moreover, dehumidification system **1700** may include one or more additional fans positioned within any one or more of these airflow paths. Similarly, while the fan **1752** of external condenser unit **1726** is depicted as being located adjacent alternate condenser **1724**, it should be understood that fan **1752** may be located anywhere (e.g., above, below, beside) with respect to alternate condenser **1724**, so long as fan **1752** is appropriately positioned and configured to facilitate flow of first outdoor airflow **1748** towards alternate condenser **1724**.

[0191] Primary metering device **1716** and secondary metering device **1718** are any appropriate type of metering/expansion device. In some embodiments, primary metering device **1716** is a thermostatic expansion valve (TXV) and secondary metering device **1718** is a fixed orifice device (or vice versa). In certain embodiments, metering devices **1716**, **1718** remove pressure from flow of refrigerant **1732** to allow expansion or change of state from a liquid to a vapor in evaporators **1710**, **1706**. The high-pressure liquid (or mostly liquid) refrigerant entering metering devices **1716**, **1718** is at a higher temperature than the liquid refrigerant **1732** leaving metering devices **1716**, **1718**. For example, if flow of refrigerant **1732** entering primary metering device **1716** is 340 psig/80° F./0% vapor, flow of refrigerant **1732** may be 196 psig/68° F./5% vapor as it leaves

primary metering device **1716**. As another example, if flow of refrigerant **1732** entering secondary metering device **1718** is 196 psig/68° F./4% vapor, flow of refrigerant **1732** may be 128 psig/44° F./14% vapor as it leaves secondary metering device **1718**.

[0192] Refrigerant **1732** may be any suitable refrigerant such as R410a. In general, dehumidification system **1700** utilizes a closed refrigeration loop of refrigerant **1732** that passes from compressor **1714** through second modulating valve **1722**, primary condenser **1708** and/or alternate condenser **1724**, (optionally) sub-cooling coil **1728**, primary metering device **1716**, secondary evaporator **1710**, first modulating valve **1702**, secondary condenser **1712** and/or secondary metering device **1718** (where refrigerant **1732** may bypass secondary condenser **1712**), primary evaporator **1704**, and superheat control evaporator **1704**. Compressor **1714** pressurizes flow of refrigerant **1732**, thereby increasing the temperature of refrigerant **1732**. Primary and secondary condensers **1708**, **1712**, which may include any suitable heat exchangers, cool the pressurized flow of refrigerant **1732** by facilitating heat transfer from the flow of refrigerant **1732** to the respective airflows passing through them (i.e., fourth or fifth airflow **1744**, **1746** and third airflow **1742**, respectively). Further, alternate condenser **1724**, which may include any suitable heat exchanger, cools the pressurized flow of refrigerant **1732** by facilitating heat transfer from the flow of refrigerant **1732** to either the airflow passing through it (i.e., first outdoor airflow **1748**) or to the flow of a fluid provided by an external source. The cooled flow of refrigerant **1732** leaving primary and/or alternate condensers **1708**, **1724** may enter primary metering device **1716**, which is operable to reduce the pressure of flow of refrigerant **1732**, thereby reducing the temperature of flow of refrigerant **1732**. In embodiments, the refrigerant **1732** may first flow through the optional sub-cooling coil **1728** before engaging with the primary metering device **1716**. Depending on the mode of operation, the cooled flow of refrigerant **1732** leaving secondary condenser **1712** may enter secondary metering device **1718**, which is operable to reduce the pressure of flow of refrigerant **1732**, thereby reducing the temperature of flow of refrigerant **1732**. The refrigerant **1732** may alternatively be received by the secondary metering device **1718** from the first modulating valve **1702**, bypassing the secondary condenser **1712**. Primary and secondary evaporators **1706**, **1710**, which may include any suitable heat exchanger, receive flow of refrigerant **1732** from secondary metering device **1718** and primary metering device **1716**, respectively. Primary and secondary evaporators **1706**, **1710** facilitate the transfer of heat from the respective airflows passing through them to the flow of refrigerant **1732**. Flow of refrigerant **1732**, after leaving primary evaporator **1706**, may be received by the superheat control evaporator **1704**. The superheat control evaporator **1704** may facilitate transfer of heat from inlet airflow **1734** passing therethrough to the flow of refrigerant **1732**. Then, the refrigerant **1732** may be directed back to compressor **1714**, and the cycle is repeated.

[0193] In certain embodiments, the above-described refrigeration loop may be configured such that evaporators **1706**, **1710** operate in a flooded state. In other words, flow of refrigerant **1732** may enter evaporators **1706**, **1710** in a liquid state, and a portion of flow of refrigerant **1732** may still be in a liquid state as it exits evaporators **1706**, **1710**. Accordingly, the phase change of flow of refrigerant **1732** (liquid to vapor as heat is transferred to flow of refrigerant **1732**) occurs across evaporators **1706**, **1710**, resulting in nearly constant pressure and temperature across the entire evaporators **1706**, **1710** (and, as a result, increased cooling capacity). In these embodiments, superheat control evaporator **1704** may additionally operate in a flooded state.

[0194] In operation of example embodiments of dehumidification system **1700**, one or more inlet airflows **1734** may be drawn into dehumidification system **1700** by fan **1720**. The one or more inlet airflows **1734** may pass through superheat control evaporator **1704** and/or secondary evaporator **1710** in which heat is transferred from the one or more inlet airflows **1734** to the cooler flow of refrigerant **1732** passing through evaporators **1704**, **1710**. As a result, the one or more inlet airflows **1734** may be cooled. As an example, if one or more inlet airflows **1734** is 80° F./60% humidity, superheat control evaporator **1704** and/or secondary evaporator **1710** may output first airflow **1738**

and/or second airflow **1740** at 70° F./84% humidity. This may cause flow of refrigerant **1732** to partially vaporize within superheat control evaporator **1704** and/or secondary evaporator **1710**. For example, if flow of refrigerant **1732** entering superheat control evaporator **1704** and/or secondary evaporator **1710** is 196 psig/68° F./5% vapor, flow of refrigerant **1732** may be 196 psig/68° F./38% vapor as it leaves superheat control evaporator **1704** and/or secondary evaporator **1710**.

[0195] The cooled one or more inlet airflows **1734** may be discharged from superheat control evaporator **1704** and/or secondary evaporator **1710** as first airflow **1738** and/or second airflow, respectively, and may enter primary evaporator **1706**. Like superheat control evaporator **1704** and/or secondary evaporator **1710**, primary evaporator **1706** may transfer heat from first airflow **1738** and/or second airflow **1740** to the cool flow of refrigerant **1732** passing through primary evaporator **1706**. As a result, the air may be cooled to or below its dew point temperature, causing moisture in first airflow **1738** and/or second airflow **1740** to condense (thereby reducing the absolute humidity of first airflow **1738** and/or second airflow **1740**). As an example, if first airflow **1738** and/or second airflow **1740** is 70° F./84% humidity, primary evaporator **1706** may output third airflow **1742** at 54° F./98% humidity. This may cause the flow of refrigerant **1732** to partially or completely vaporize within primary evaporator **1706**. For example, if flow of refrigerant **1732** entering primary evaporator **1706** is 128 psig/44° F./14% vapor, flow of refrigerant **1732** may be 128 psig/52° F./100% vapor as it leaves primary evaporator **1706**.

[0196] The third airflow **1742** may be discharged from primary evaporator **1706** and may enter secondary condenser **1712**. Secondary condenser **1712** may be configured to facilitate heat transfer from the hot flow of refrigerant **1732** passing through the secondary condenser **1712** to third airflow **1742**, depending on the mode of operation. This reheats third airflow **1742**, thereby decreasing the relative humidity of third airflow **1742**. As an example, if third airflow **1742** is 54° F./98% humidity, secondary condenser **1712** may output fourth airflow **1744** at 65° F./68% humidity. This may cause flow of refrigerant **1732** to partially or completely condense within secondary condenser **1712**. For example, if flow of refrigerant **1732** entering secondary condenser **1712** is 196 psig/68° F./38% vapor, flow of refrigerant **1732** may be 196 psig/68° F./4% vapor as it leaves secondary condenser **1712**.

[0197] In some embodiments, the fourth airflow **1744** may be discharged and may enter the optional sub-cooling coil **1728**. Sub-cooling coil **1728** facilitates heat transfer from the hot flow of refrigerant **1732** passing through sub-cooling coil **1728** to fourth airflow **1744** to generate the fifth airflow **1746** to be output to the primary condenser **1708**. In other embodiments, the fourth airflow **1744** may be discharged and may enter the primary condenser **1708** without flowing through the sub-cooling coil **1728**. Primary condenser **1708** facilitates heat transfer from the hot flow of refrigerant **1732** passing through the primary condenser **1708** to fourth airflow **1744** or fifth airflow **1746**. This further heats fourth airflow **1744** or fifth airflow **1746**, thereby further decreasing the relative humidity of fourth airflow **1744** or fifth airflow **1746**. As an example, if fourth airflow **1744** or fifth airflow **1746** is 65° F./68% humidity, primary condenser **1708** may output dischargeable airflow **1736** at 102° F./19% humidity. This may cause flow of refrigerant **1732** to partially or completely condense within primary condenser **1708**. For example, if flow of refrigerant **1732** entering primary condenser **1708** is 340 psig/150° F./100% vapor, flow of refrigerant **1732** may be 340 psig/105° F./60% vapor as it leaves primary condenser **1708**.

[0198] Some embodiments of dehumidification system **1700** may include a controller that may include one or more computer systems at one or more locations. Each computer system may include any appropriate input devices (such as a keypad, touch screen, mouse, or other device that can accept information), output devices, mass storage media, or other suitable components for receiving, processing, storing, and communicating data. Both the input devices and output devices may include fixed or removable storage media such as a magnetic computer disk, CD-ROM, or other suitable media to both receive input from and provide output to a user. Each computer system may include a personal computer, workstation, network computer, kiosk, wireless data port,

personal data assistant (PDA), one or more processors within these or other devices, or any other suitable processing device. In short, the controller may include any suitable combination of software, firmware, and hardware.

[0199] The controller may additionally include one or more processing modules. Each processing module may each include one or more microprocessors, controllers, or any other suitable computing devices or resources and may work, either alone or with other components of dehumidification system **1700**, to provide a portion or all of the functionality described herein. The controller may additionally include (or be communicatively coupled to via wireless or wireline communication) computer memory. The memory may include any memory or database module and may take the form of volatile or non-volatile memory, including, without limitation, magnetic media, optical media, random access memory (RAM), read-only memory (ROM), removable media, or any other suitable local or remote memory component.

[0200] Although particular implementations of dehumidification system **1700** are illustrated and primarily described, the present disclosure contemplates any suitable implementation of dehumidification system **1700**, according to particular needs. Moreover, although various components of dehumidification system **1700** have been depicted as being located at particular positions and relative to one another, the present disclosure contemplates those components being positioned at any suitable location, according to particular needs.

[0201] Herein, a computer-readable non-transitory storage medium or media may include one or more semiconductor-based or other integrated circuits (ICs) (such, as for example, field-programmable gate arrays (FPGAs) or application-specific ICs (ASICs)), hard disk drives (HDDs), hybrid hard drives (HHDs), optical discs, optical disc drives (ODDs), magneto-optical discs, magneto-optical drives, floppy diskettes, floppy disk drives (FDDs), magnetic tapes, solid-state drives (SSDs), RAM-drives, SECURE DIGITAL cards or drives, any other suitable computer-readable non-transitory storage media, or any suitable combination of two or more of these, where appropriate. A computer-readable non-transitory storage medium may be volatile, non-volatile, or a combination of volatile and non-volatile, where appropriate.

[0202] Herein, “or” is inclusive and not exclusive, unless expressly indicated otherwise or indicated otherwise by context. Therefore, herein, “A or B” means “A, B, or both,” unless expressly indicated otherwise or indicated otherwise by context. Moreover, “and” is both joint and several, unless expressly indicated otherwise or indicated otherwise by context. Therefore, herein, “A and B” means “A and B, jointly or severally,” unless expressly indicated otherwise or indicated otherwise by context.

[0203] The scope of this disclosure encompasses all changes, substitutions, variations, alterations, and modifications to the example embodiments described or illustrated herein that a person having ordinary skill in the art would comprehend. The scope of this disclosure is not limited to the example embodiments described or illustrated herein. Moreover, although this disclosure describes and illustrates respective embodiments herein as including particular components, elements, feature, functions, operations, or steps, any of these embodiments may include any combination or permutation of any of the components, elements, features, functions, operations, or steps described or illustrated anywhere herein that a person having ordinary skill in the art would comprehend. Furthermore, reference in the appended claims to an apparatus or system or a component of an apparatus or system being adapted to, arranged to, capable of, configured to, enabled to, operable to, or operative to perform a particular function encompasses that apparatus, system, component, whether or not it or that particular function is activated, turned on, or unlocked, as long as that apparatus, system, or component is so adapted, arranged, capable, configured, enabled, operable, or operative. Additionally, although this disclosure describes or illustrates particular embodiments as providing particular advantages, particular embodiments may provide none, some, or all of these advantages.

Claims

1. A dehumidification system comprising: a primary metering device; a secondary metering device; an intermixed coil unit operable to: receive an inlet airflow and output a first airflow, the first airflow comprising cooler air than the first inlet airflow, the first airflow generated by transferring heat from the inlet airflow to a flow of refrigerant within a superheat control evaporator and a secondary evaporator as the inlet airflow passes through both the superheat control evaporator and the secondary evaporator, wherein the intermixed coil unit comprises: the superheat control evaporator operable to receive a flow of refrigerant from a primary evaporator; and the secondary evaporator operable to receive the flow of refrigerant from the primary metering device; a primary evaporator operable to: receive the flow of refrigerant from a first modulating valve and/or a secondary condenser; and receive the first airflow and output a second airflow, the second airflow comprising cooler air than the first airflow, the second airflow generated by transferring heat from the first airflow to the flow of refrigerant as the first airflow pass through the primary evaporator; the secondary condenser operable to: receive the flow of refrigerant from the first modulating valve; and receive the second airflow and output a third airflow; the first modulating valve operable to: receive the flow of refrigerant from the secondary evaporator; direct the flow of refrigerant to the secondary condenser during a first mode of operation; direct the flow of refrigerant to the primary evaporator during a second mode of operation, wherein the flow of refrigerant bypasses the secondary condenser; and direct a portion of the flow of refrigerant to the secondary condenser and a remaining portion of the flow of refrigerant to the primary evaporator during a third mode of operation; a compressor configured to: receive the flow of refrigerant from the superheat control evaporator and discharge the flow of refrigerant at a higher pressure than the flow of refrigerant received at the compressor; and a primary condenser operable to: receive the flow of refrigerant discharged from the compressor; in response to receiving the flow of refrigerant from the compressor, output a dischargeable airflow.
2. The dehumidification system of claim 1, further comprising a sub-cooling coil operable to: receive the flow of refrigerant from the primary condenser; output the flow of refrigerant to the primary metering device; and receive the third airflow and output a fourth airflow, the fourth airflow generated by transferring heat from the flow of refrigerant to the third airflow as the third airflow passes through the sub-cooling coil.
3. The dehumidification system of claim 2, wherein the sub-cooling coil and primary condenser are combined in a single coil unit.
4. The dehumidification system of claim 2, wherein the primary condenser is further operable to: receive the fourth airflow and generate the dischargeable airflow based on the received fourth airflow, the dischargeable airflow generated by transferring heat from the flow of refrigerant to the fourth airflow as the fourth airflow contacts the primary condenser.
5. The dehumidification system of claim 1, wherein the primary condenser is further operable to: receive the third airflow and generate the dischargeable airflow based on the received third airflow, the dischargeable airflow generated by transferring heat from the flow of refrigerant to the third airflow as the third airflow contacts the primary condenser.
6. The dehumidification system of claim 1, wherein during the first mode of operation: the third airflow comprises warmer and less humid air than the second airflow, the third airflow generated by transferring heat from the flow of refrigerant to the second airflow as the second airflow passes through the secondary condenser.
7. The dehumidification system of claim 1, wherein during the second mode of operation, the third airflow comprises approximately the same temperature and humidity as the second airflow.
8. The dehumidification system of claim 1, further comprising an alternate condenser disposed in an external condenser unit, operable to: receive the flow of refrigerant discharged from the

compressor; in response to receiving the flow of refrigerant from the compressor, output a second dischargeable airflow, the second dischargeable airflow generated by transferring heat away from the flow of refrigerant received by the compressor.

9. The dehumidification system of claim 1, wherein two or more members selected from the group consisting of the secondary evaporator, the primary evaporator, and the secondary condenser are combined in a single coil pack.

10. The dehumidification system of claim 1, wherein at least one of the primary evaporator and the secondary evaporator comprises two or more circuits for the flow of refrigerant.
